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**Wong et al.**

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(54) **ADDITIVE MANUFACTURING FOR MEDICAL DEVICES**

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**B29C 64/118** (2017.01)  
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(58) **Field of Classification Search**  
None  
See application file for complete search history.

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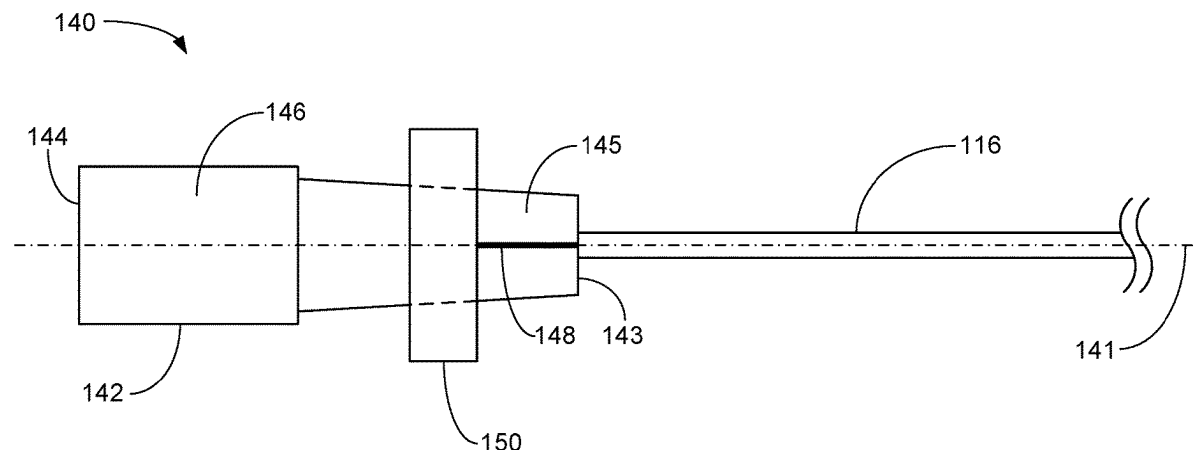
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(57) **ABSTRACT**

An additive manufacturing system for producing a medical catheter or lead and a method thereof. The system including a heating cartridge defining an interior volume and at least one filament port. The system also including a heating element thermally coupled to the heating cartridge to heat the interior volume, a filament handling system to feed at least one filament through the at least one filament port, and a substrate handling system. The substrate handling system including a clamp to secure a portion of a substrate to be moved relative to the heating cartridge to apply a jacket to the substrate. In one or more embodiments, a subassembly is positioned on the substrate and has an electrode ring. The jacket printed to cover at least a portion of the subassembly and spaced apart from the electrode ring.

**21 Claims, 14 Drawing Sheets**



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*B29L 31/00* (2006.01)  
*B33Y 10/00* (2015.01)  
*B33Y 30/00* (2015.01)  
*B33Y 80/00* (2015.01)

(52) **U.S. Cl.**

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(2014.12); *B33Y 80/00* (2014.12)

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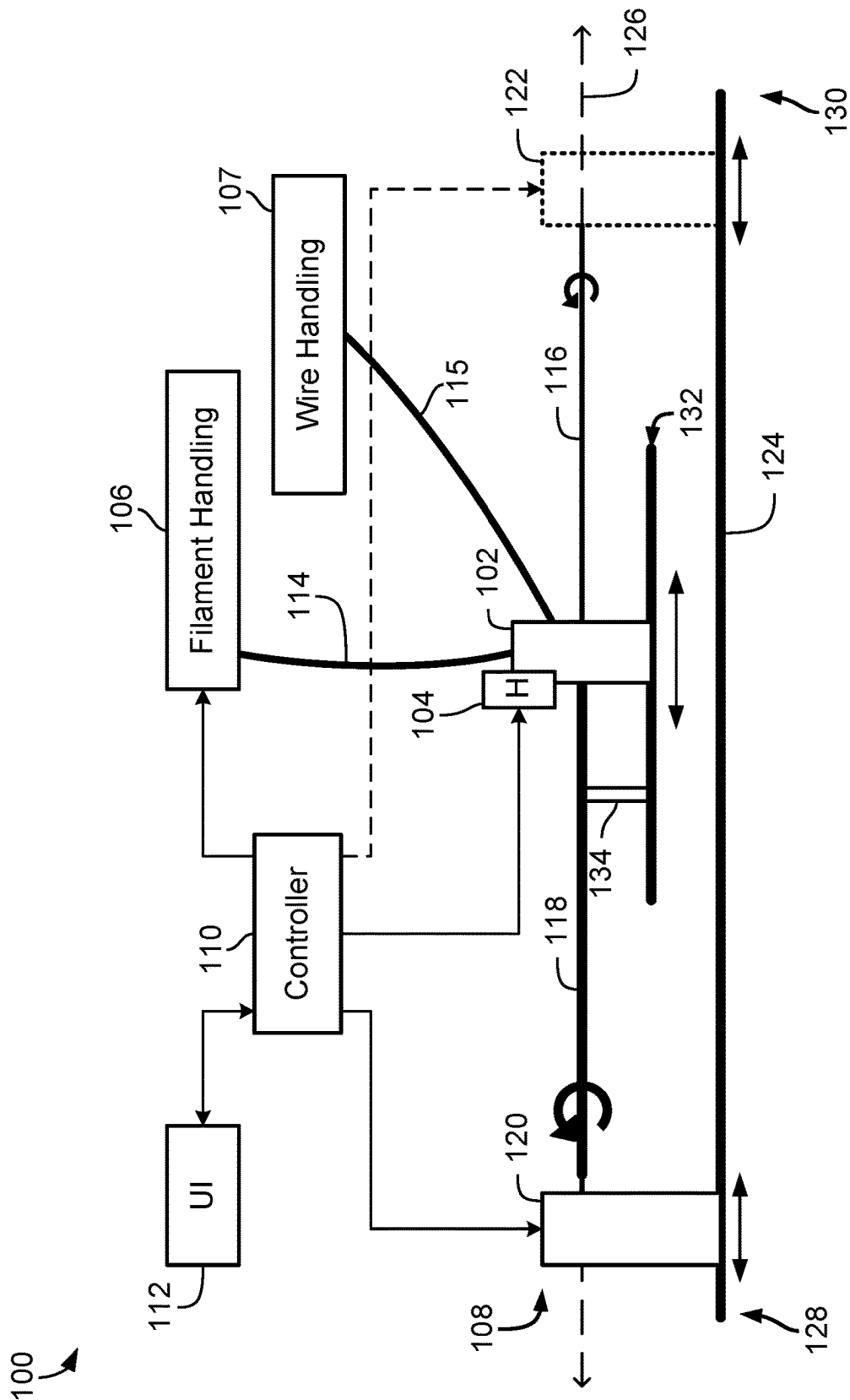
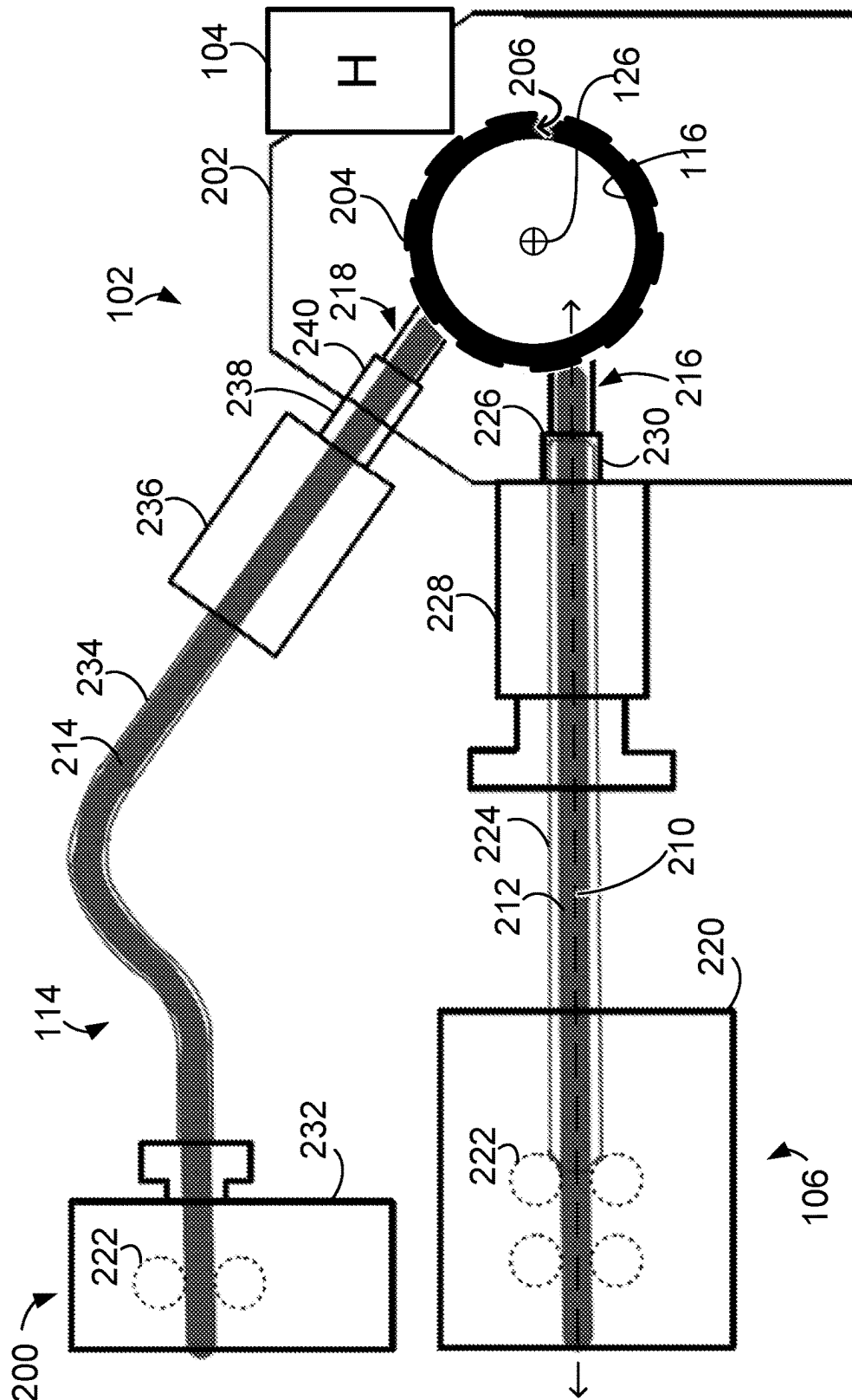


FIG. 1



**FIG. 2**

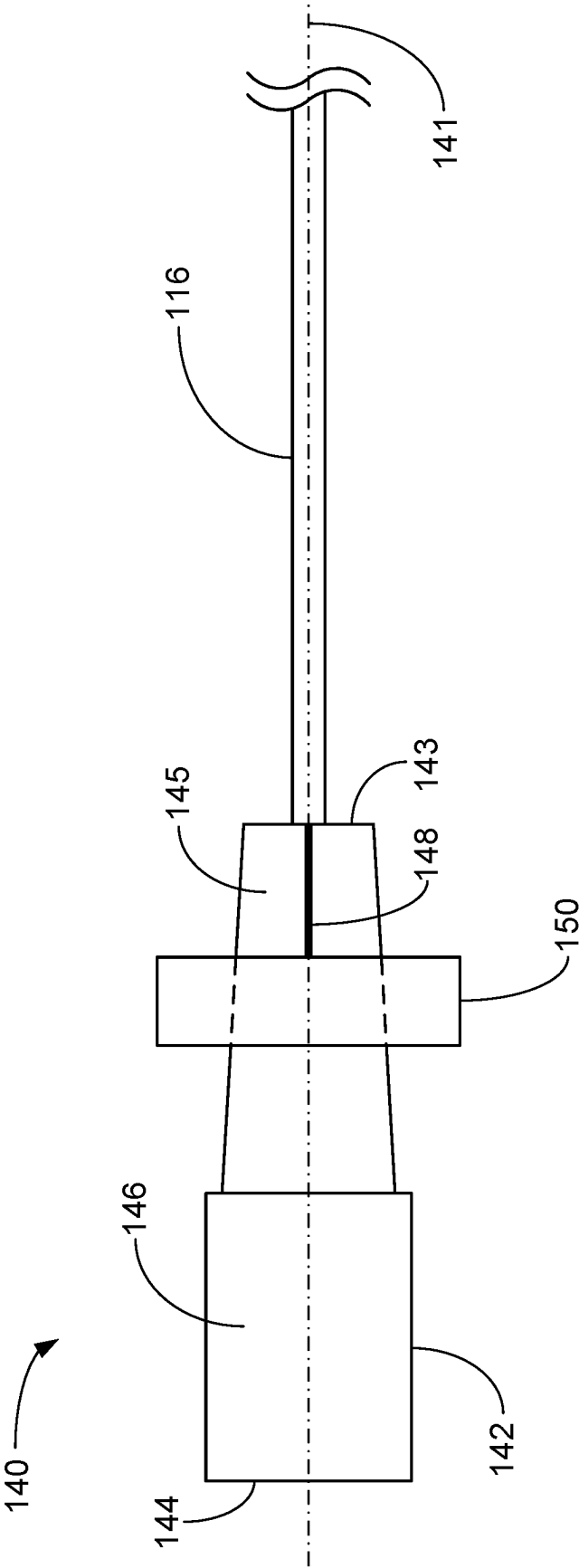


FIG. 3

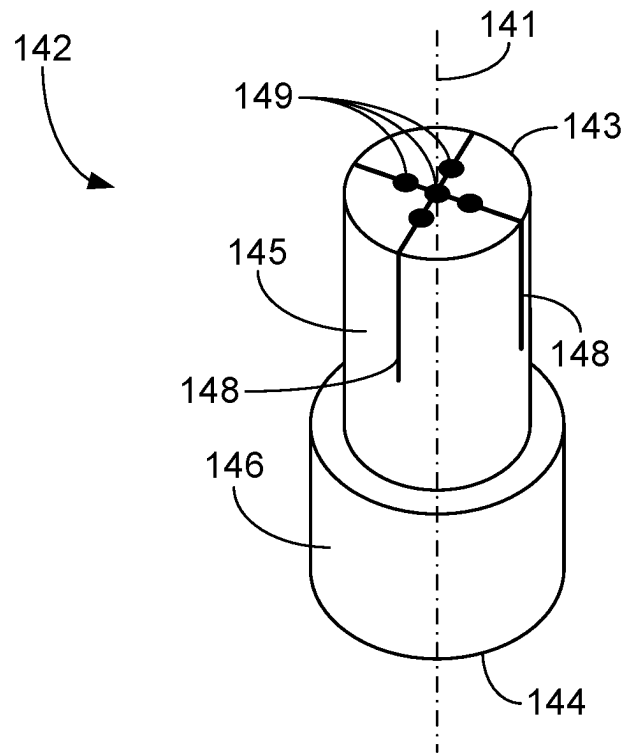


FIG. 4A

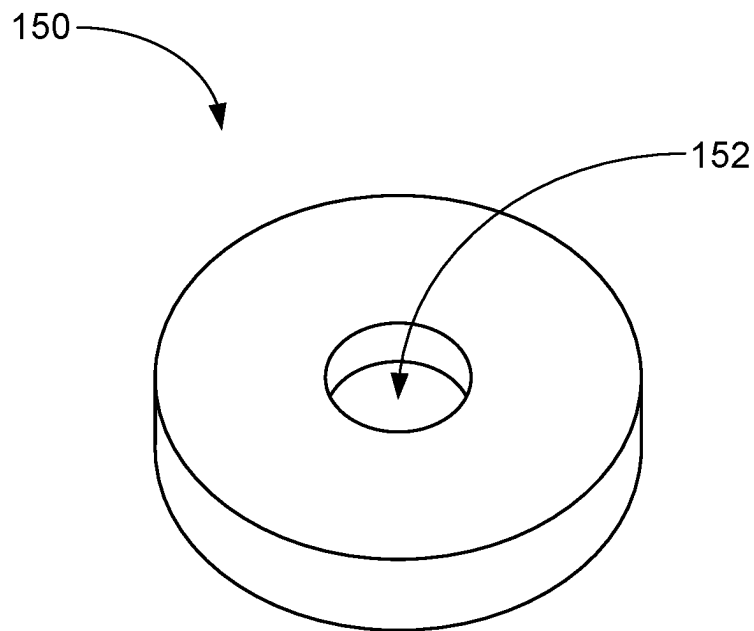


FIG. 4B

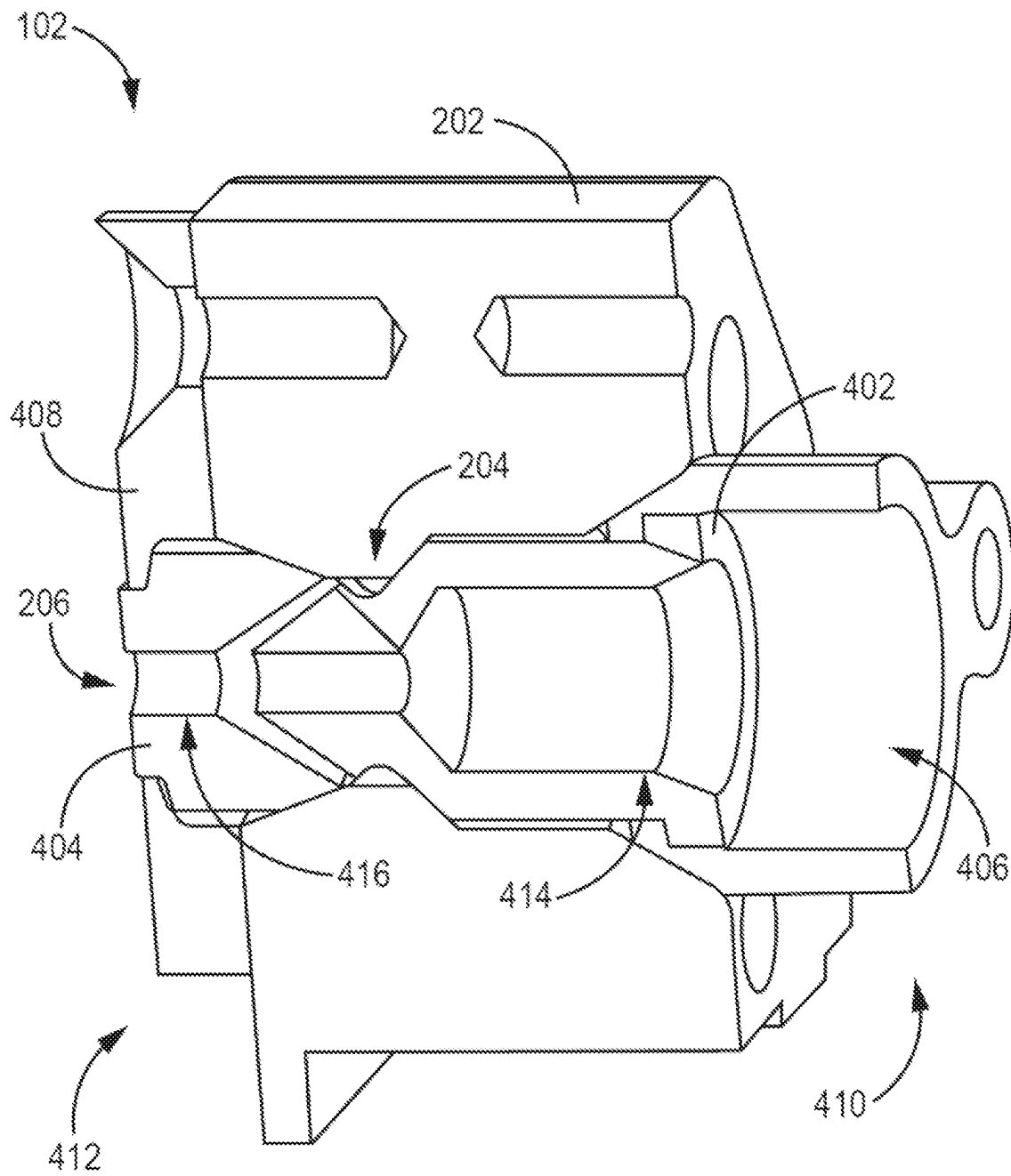


FIG. 5

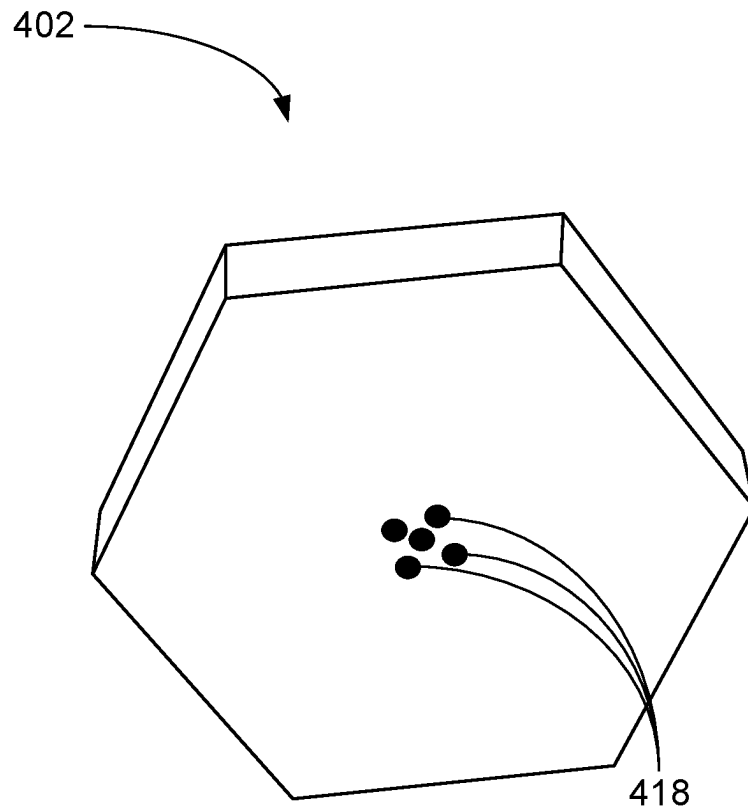


FIG. 6



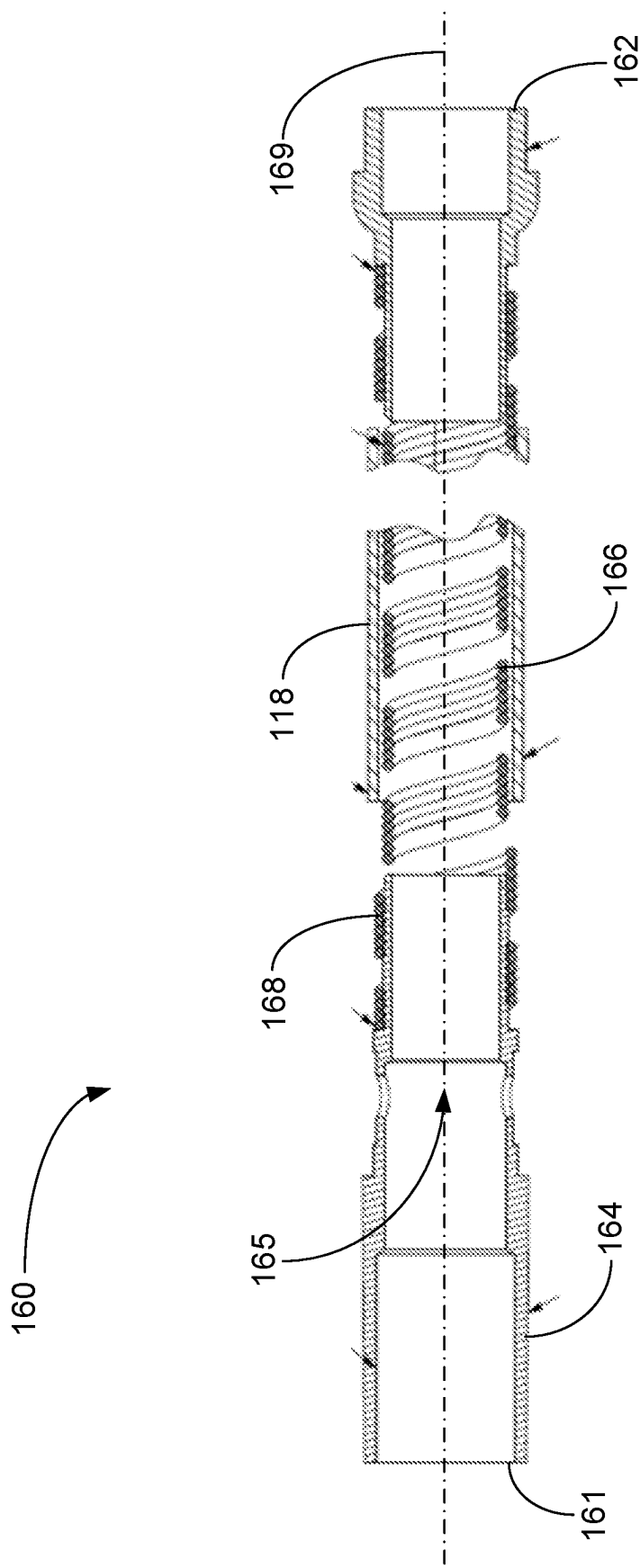


FIG. 7

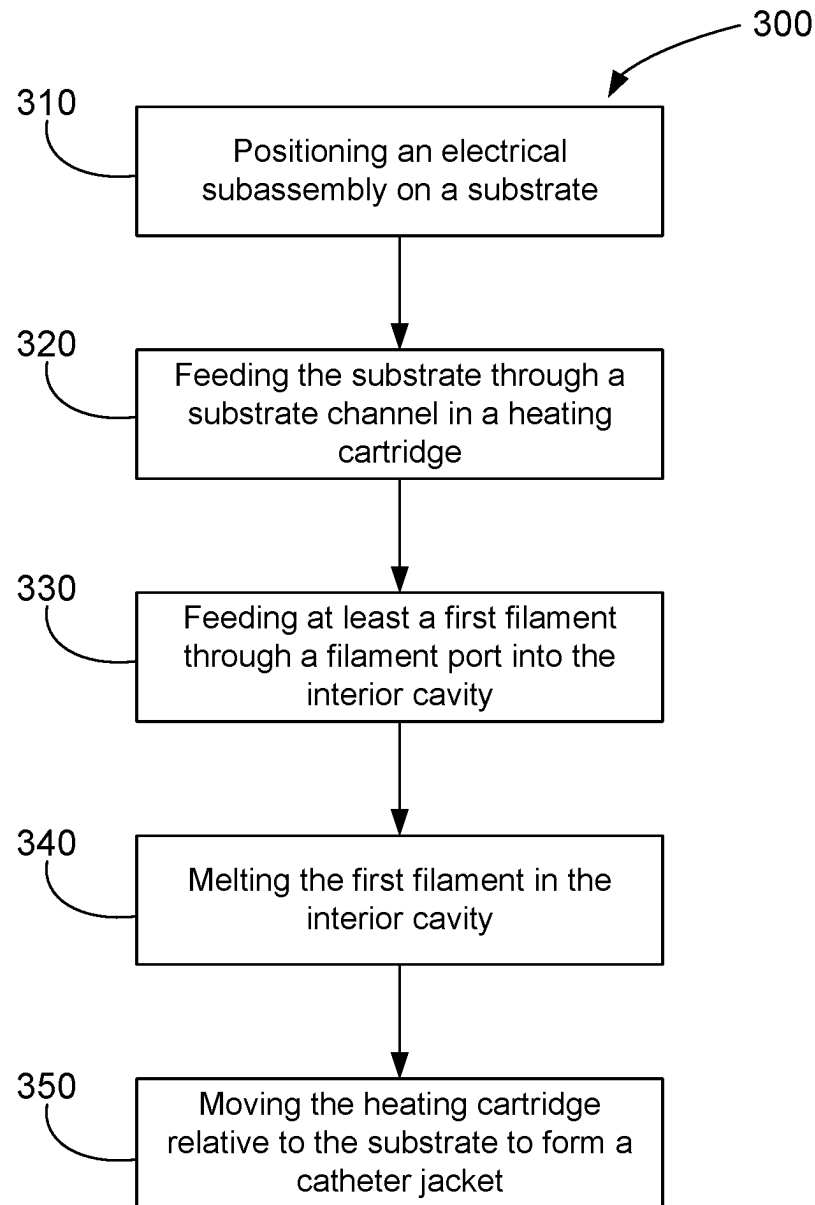


FIG. 8

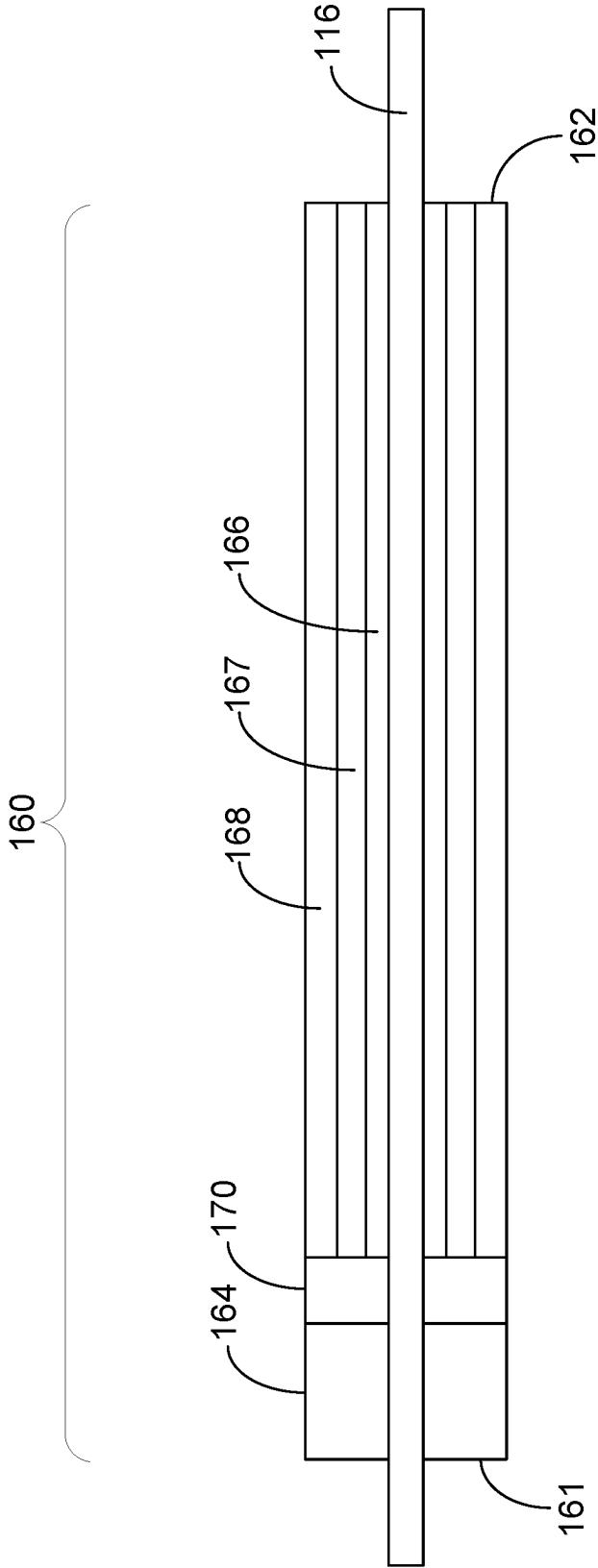


FIG. 9A

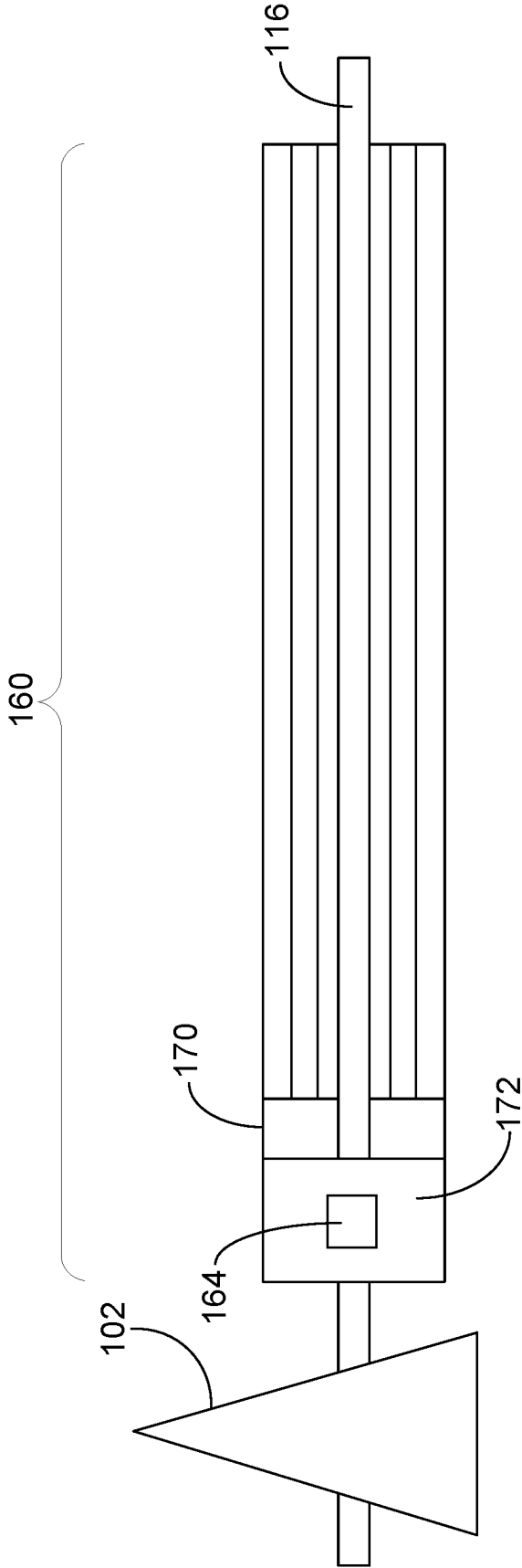


FIG. 9B

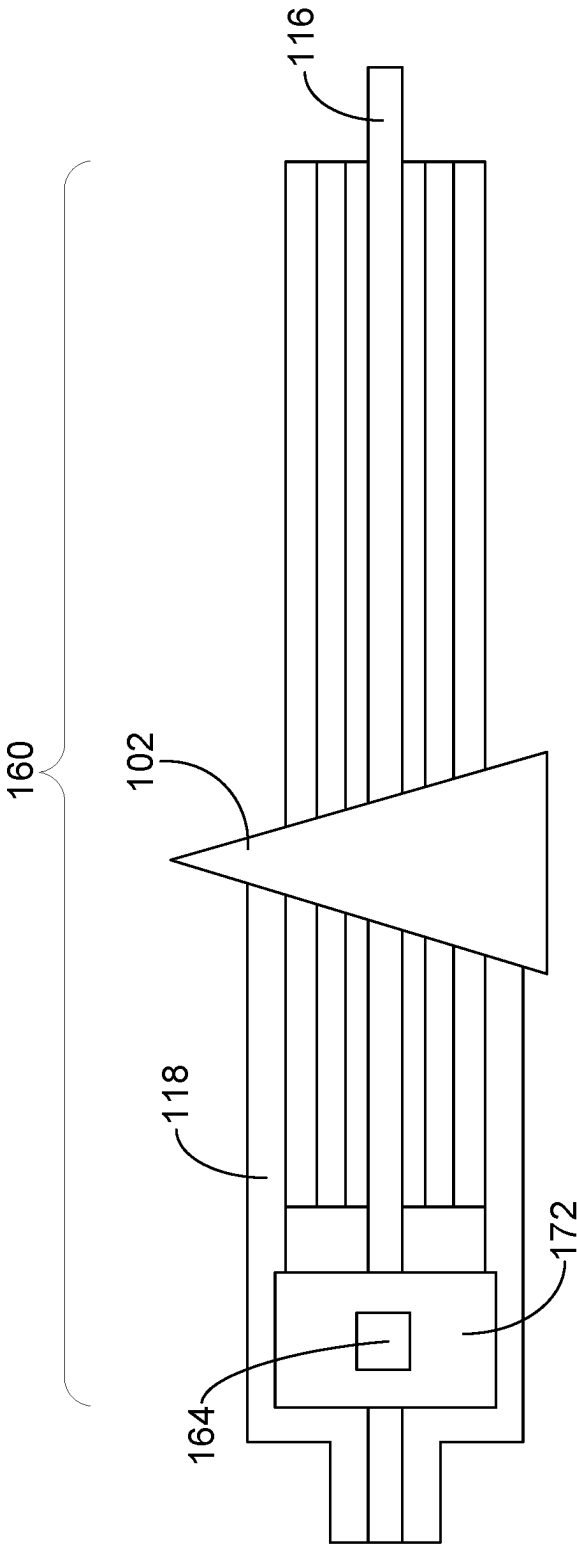


FIG. 9C

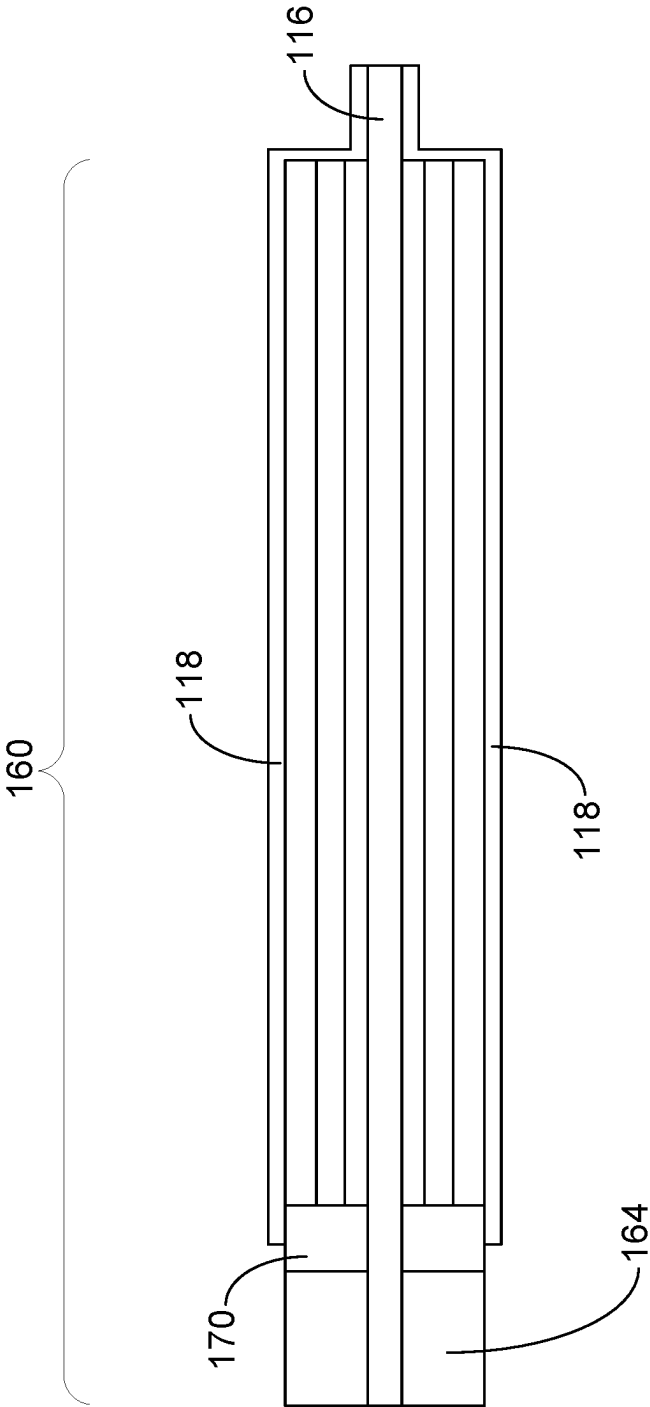


FIG. 9D

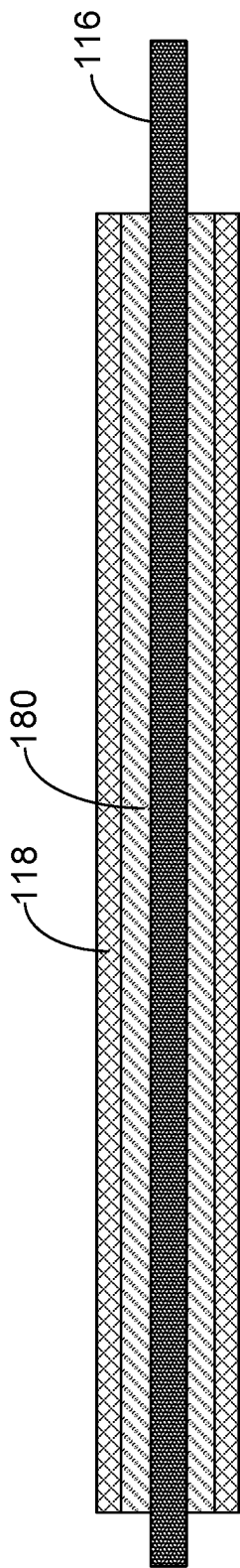


FIG. 10

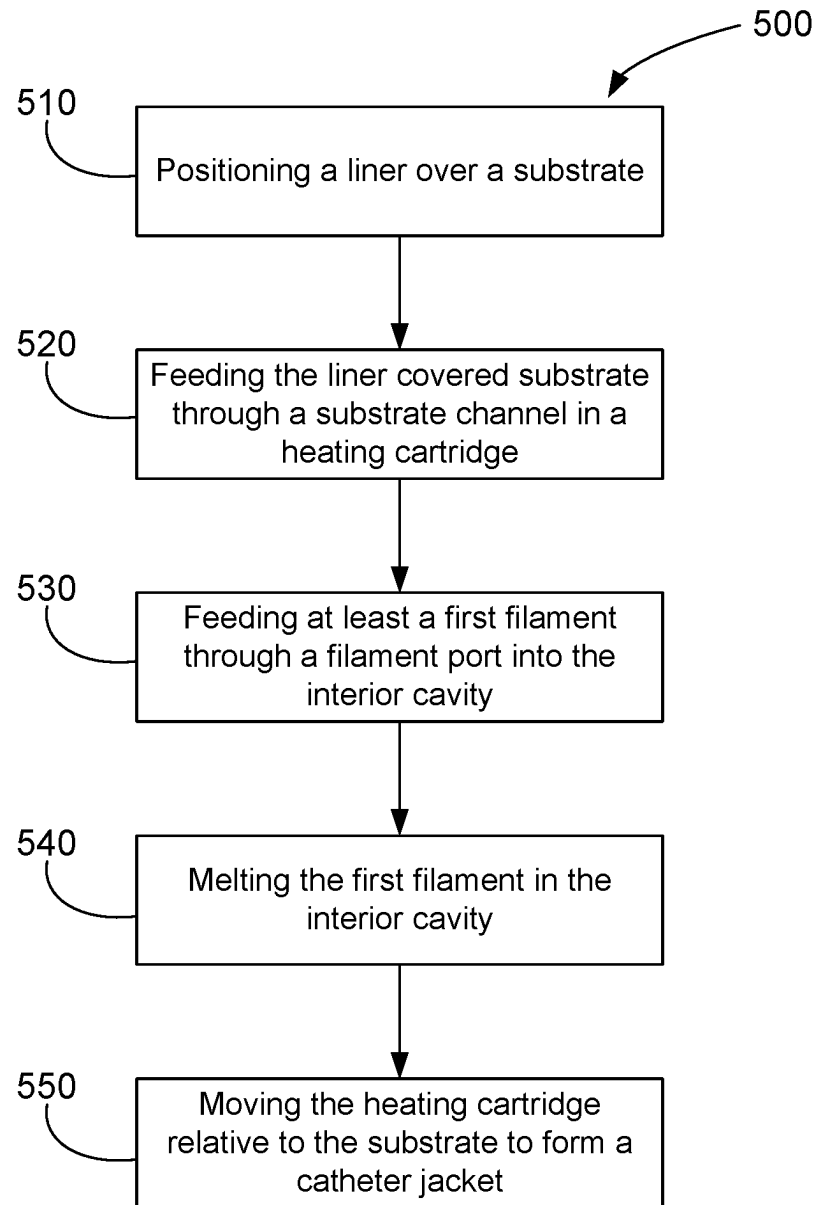


FIG. 11



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## ADDITIVE MANUFACTURING FOR MEDICAL DEVICES

### RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application Ser. No. 63/388,049, filed Jul. 11, 2022, the entire contents of each of which are incorporated herein by reference. The disclosure generally relates to medical devices and, in particular, additive manufacturing of medical devices, such as catheters and implantable stimulation leads.

Medical catheters and leads are commonly used to access vascular and other locations within a body and to perform various functions at those locations, for example, delivery catheters may be used to deliver medical devices, such as implantable medical leads. A number of such medical devices are designed to be navigated through tortuous paths in a human body, such as through a patient's vasculature. Medical catheters and leads may be designed to be sufficiently flexible to move through turns, or curves, in the vasculature yet sufficiently stiff, or resilient, to be pushed through the vasculature. In many cases, such as those involving cardiovascular vessels, the route to the treatment or deployment site may be tortuous and may present conflicting design considerations that may require compromises between dimensions, flexibilities, material selection, operational controls and the like. These contrasting properties can present challenges in designing and manufacturing catheters. Existing manufacturing processes, such as conventional extrusion, may also limit options in designing and manufacturing catheters.

### SUMMARY

The techniques of the present disclosure generally relate to additive manufacturing of medical devices, such as catheters and leads, that allows for further customization of the medical devices by/providing an easier way to include components internal to the medical device. For example, a subassembly may be positioned on a substrate or mandrel and a layer printed thereover, thereby encapsulating the components located in the subassembly. These systems and techniques may allow for manufacturing more complex medical devices without increasing the complexity of manufacturing. Further, the systems and techniques described herein may include positioning a liner on the substrate or mandrel prior to printing such that the substrate or mandrel can easily move relative to the printed jacket. As such, in some embodiments, the substrate or mandrel may include a lead such that the jacket is printed around the lead and the lead still has flexibility relative to the jacket.

Additionally, the additive manufacturing systems may include components for holding the one or more substrates in position for printing thereon. For example, an inlet die of a heating cartridge through which the substrate passes may define opening(s) to guide the substrate(s) in position. Further, the system may include a clamp to hold the substrate(s) in position at an end of the substrate(s). Specifically, in one or more embodiments, there may be multiple substrates such that the inlet die and the clamp include a corresponding number of openings to receive the substrates.

The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the tech-

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niques described in this disclosure will be apparent from the description and drawings, and from the claims.

### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a conceptual diagram that illustrates one example of an additive manufacturing system according to the present disclosure.

FIG. 2 is a conceptual diagram that illustrates one example of an additive manufacturing apparatus for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 3 is a conceptual diagram that illustrates one example of a clamp attached to a substrate for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 4A is a perspective view of one example of a sleeve of the illustrative clamp of FIG. 3.

FIG. 4B is a perspective view of one example of a collar of the illustrative clamp of FIG. 3.

FIG. 5 is a conceptual diagram that illustrates one example of a heating cartridge for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 6 is a conceptual diagram that illustrates one example of an inlet die of a heating cartridge for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 7 is a cross-sectional view of an illustrative subassembly for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 8 is a flow diagram that illustrates one example of a method for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 9A is a conceptual diagram that illustrates a subassembly positioned on a substrate for use with, for example, the additive manufacturing system of FIG. 1.

FIG. 9B is a conceptual diagram of the subassembly and substrate of FIG. 9A positioned relative to a heating cartridge.

FIG. 9C is a conceptual diagram of the subassembly and substrate of FIG. 9B passing through the heating cartridge to print a jacket thereon.

FIG. 9D is a conceptual diagram of the subassembly and substrate of FIG. 9C with the jacket disposed thereon and an electrode ring exposed.

FIG. 10 is a conceptual diagram that illustrates one example of a catheter that may be manufactured with a liner using, for example, the additive manufacturing system of FIG. 1.

FIG. 11 is a flow diagram that illustrates one example of a method of manufacturing the catheter of FIG. 10 using, for example, the additive manufacturing system of FIG. 1.

### DETAILED DESCRIPTION

The present disclosure generally provides additive manufacturing systems and methods for medical devices, such as catheters and leads, that allows for customization of the medical devices by using subassemblies and liners. For example, a jacket (e.g., a catheter jacket or insulation) may be printed over a subassembly located on a substrate/mandrel. It is noted that catheter jacket and printed jacket may be used interchangeably herein. The subassembly may include a variety of different components that can be embedded in the medical device by the printed jacket. For example, the subassembly may include electrical components such as, e.g., an electrode ring, coils, wires, etc. and the jacket may be printed over the subassembly to embed those components. Although, in some embodiments, the electrode ring

may be left exposed (or any material printed on the electrode ring may be removed) to provide an operable electrode ring on the medical device.

Further, in one or more embodiments, the substrate/mandrel may include a liner (e.g., a harvestable liner) positioned over or covering the substrate/mandrel. Therefore, the jacket may be printed on the liner and the substrate/mandrel may be easily removed from the liner (e.g., leaving the printed jacket to form the medical device having a lumen extending therethrough). In one or more embodiments, the substrate/mandrel may be replaced by a lead or wire such that the jacket is printed over the lead or wire (e.g., with the liner positioned therebetween). Because the liner is present between the printed jacket and the lead/wire, the lead/wire may have flexibility within the printed jacket (e.g., the lead/wire may move relative to the printed jacket). In other words, the printed jacket may not be rigidly adhered to the lead or wire in such a way that the flexibility of the lead or wire is impeded thereby.

Further, the additive manufacturing systems may include components to assist in holding and guiding one or multiple substrates/mandrels. For example, a heating cartridge of the system may include an inlet die and the inlet die may be configured to help guide the substrates into and through the heating cartridge. Specifically, the inlet die may include one or more openings that correspond to the desired configuration of the substrates (e.g., upon which material may be printed to form a jacket). In other words, the configuration of the inlet die may define the internal structure of the medical device.

Additionally, the system may include a head stock to hold the substrate opposite the inlet die (e.g., such that the substrate extends between the inlet die and the head stock) and assist in moving the substrate. The head stock may include a clamp also having one or more openings that correspond to the desired configuration of the substrates. Further, the clamp may fix the substrate in the system such that the substrate may move relative to the heating cartridge. Specifically, the clamp may include a sleeve to receive the substrate and a collar to compress the sleeve onto the substrate (e.g., to restrict movement of the substrate). In other words, the collar may be moved relative to the sleeve to deform at least a portion of the sleeve such that the sleeve restricts movement of the substrate.

As used herein, the term “or” refers to an inclusive definition, for example, to mean “and/or” unless its context of usage clearly dictates otherwise. The term “and/or” refers to one or all of the listed elements or a combination of at least two of the listed elements.

As used herein, the phrases “at least one of” and “one or more of” followed by a list of elements refers to one or more of any of the elements listed or any combination of one or more of the elements listed.

As used herein, the terms “coupled” or “connected” refer to at least two elements being attached to each other either directly or indirectly. An indirect coupling may include one or more other elements between the at least two elements being attached. Further, in one or more embodiments, one element “on” another element may be directly or indirectly on and may include intermediate components or layers therebetween. Either term may be modified by “operatively” and “operably,” which may be used interchangeably, to describe that the coupling or connection is configured to allow the components to interact to carry out described or otherwise known functionality. For example, a controller

may be operably coupled to a resistive heating element to allow the controller to provide an electrical current to the heating element.

As used herein, any term related to position or orientation, such as “proximal,” “distal,” “end,” “outer,” “inner,” and the like, refers to a relative position and does not limit the absolute orientation of an embodiment unless its context of usage clearly dictates otherwise.

All scientific and technical terms used herein have meanings commonly used in the art unless otherwise specified. The definitions provided herein are to facilitate understanding of certain terms used frequently herein and are not meant to limit the scope of the present disclosure.

Reference will now be made to the drawings, which depict one or more aspects described in this disclosure. However, it will be understood that other aspects not depicted in the drawings fall within the scope of this disclosure. Like numbers used in the figures refer to like components, steps, and the like. However, it will be understood that the use of a reference character to refer to an element in a given figure is not intended to limit the element in another figure labeled with the same reference character. In addition, the use of different reference characters to refer to elements in different figures is not intended to indicate that the differently referenced elements cannot be the same or similar.

FIG. 1 shows one example of an additive manufacturing system **100** according to the present disclosure. The system **100** may be configured and used to create a medical device such as, e.g., a catheter, catheter component, lead, an assembly, etc. For example, consumable filament materials (e.g., having a variety of different hardness levels) may be melted to print and form components of the medical device (e.g., a printed jacket). The system **100** may be configured to operate a wide variety of process conditions to produce catheters, catheter components, leads, or assemblies using filaments or pellet to form resins of various hardness levels. In general, the system **100** defines a distal region **128**, or distal end, and a proximal region **130**, or proximal end. The system **100** may include a platform **124** including a rigid frame to support one or more components of the system.

As shown in the illustrated embodiment, the system **100** may include one or more components, such as a heating cartridge **102**, a heating element **104**, a filament handling system **106**, an optional wire handling system **107**, a substrate handling system **108**, a controller **110**, and a user interface **112**. The filament handling system **106** may be operably coupled to the heating cartridge **102**. The filament handling system **106** may provide one or more filaments **114** to the heating cartridge **102**. The optional wire handling system **107** may be used to provide one or more wires **115** to the heating cartridge **102**. The heating element **104** may be operably coupled, or thermally coupled, to the heating cartridge **102**. The heating element **104** may provide heat to melt filament material in the heating cartridge **102** from the one or more filaments **114** provided by the filament handling system **106**. The optional wires **115** may not be melted by the heating cartridge **102**. The substrate handling system **108** may be operably coupled to the heating cartridge **102**. The substrate handling system **108** may provide a substrate **116** that extends through the heating cartridge **102**. Melted filament material located in the heating cartridge **102** may be applied to the substrate **116**. The substrate **116** or the heating cartridge **102** may be moved (e.g., translated or rotated) relative to the other by the substrate handling system **108**. Specifically, the substrate handling system **108** may be used to move the substrate **116** or the heating cartridge **102**

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relative to the other to cover the substrate **116** with the melted filament material to form a jacket **118**. Further, in one or more embodiments, the optional wires **115** may be incorporated into the jacket **118** (e.g., molded into, bedded within, etc.).

The substrate **116** may also be described as a mandrel or rod. The jacket **118** may be formed or deposited around the substrate **116**. In some embodiments, the jacket **118** may be formed concentrically around the substrate **116**. In one example, the jacket **118** is formed concentrically and centered around the substrate **116**.

When the system **100** is used to make a catheter or catheter component, the jacket **118** may be described as a printed or catheter jacket. Some or all of the substrate **116** may be removed or separated from the jacket **118** and the remaining structure coupled to the jacket may form the catheter or catheter component, such as a sheath. One example of a catheter that may be formed by the system **100** is shown in FIG. **10**.

The substrate **116** may be formed of any suitable material capable of allowing melted filament material to be formed thereon. In some embodiments, the substrate **116** is formed of a material that melts at a higher temperature than any of the filaments **114**. One example of a material that may be used to form the substrate **116** includes stainless steel.

The controller **110** may be operably coupled to one or more of the heating element **104**, the filament handling system **106**, the substrate handling system **108**, and the user interface **112**. The controller **110** may activate, or initiate or otherwise “turn on,” the heating element **104** to provide heat to the heating cartridge **102** to melt the filament material located therein. Further, the controller **110** may control or command one or more motors or actuators of various portions of the system **100**. Furthermore, the controller **110** may control one or more motors or actuators of the filament handling system **106** to provide one or more filaments **114**. Further, the controller **110** may control one or more motors or actuators of the substrate handling system **108** to move one or both of the heating cartridge **102** or the substrate **116** relative to one another. Further still, the controller **110** may send or receive data to the user interface **112**, for example, to display information or to receive user commands. Control of the components operably coupled to the controller **110** may be determined based on user commands received by the user interface **112**. In some embodiments, the user commands may be provided in the form of a machine-readable code or coding language.

Any suitable implementation may be used to provide the substrate handling system **108**. In some embodiments, the substrate handling system **108** may include one or more of a head stock **120**, an optional tail stock **122**, and one or more motors coupled to or included in the head stock or tail stock. One or both of the head stock **120** and the tail stock **122** may be coupled to the platform **124**. A stock may be defined as a structure that holds or secures the substrate **116** during formation of the jacket **118**. The head stock **120** is defined as the stock closest to the end of the substrate **116** where formation of the jacket **118** begins in the formation process. In the illustrated embodiment, the jacket **118** is shown proximal to the head stock **120** and distal to the heating cartridge **102**.

When the substrate **116** is secured by one or both stocks **120**, **122**, the substrate is generally positioned to pass through a substrate channel defined by the heating cartridge **102**. One or both stocks **120**, **122** may include a clamp or other securing components to selectively hold the substrate **116**. For example, as shown in FIG. **3**, the clamp **140** may

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include a sleeve **142** and a collar **150**. The elongate substrate **116** may extend through the sleeve **142** (e.g., via an opening extending through the sleeve **142**). Further, the collar **150** may be configured to move the sleeve **142** relative to the elongate substrate **116** to, e.g., restrict movement of the substrate **116**. In other words, the collar **150** may interact with the sleeve **142** to clamp the substrate **116** (e.g., into the head stock **120**). As such, the collar **150** may engage the sleeve **142** to restrict or prevent movement of the substrate **116** and the collar **150** may disengage with the sleeve **142** such that the substrate **116** can move relative to the sleeve **142**. The clamp **140** may be included in one or both of the head stock **120** and the tail stock **122**.

The collar **150** may interact with the sleeve **142** to retain the substrate **116** in any suitable way. For example, the collar **150** may define an inner hole **152** (e.g., as shown in FIG. **4B**) to receive the sleeve **142**. The collar **150** may be configured to slide along the sleeve **142** to move at least a portion of the sleeve **142** to compress the sleeve **142** against the substrate **116** and restrict movement of the substrate **116** relative to the sleeve **142**. In other words, sliding the collar **150** onto the sleeve **142** may deform the sleeve **142** such that the sleeve **142** compresses against the substrate **116**.

Specifically, the sleeve **142** may include a deformable portion **145** and a rigid portion **146** arranged along a longitudinal axis **141** (e.g., as shown in FIG. **4A**). Further, the sleeve **142** may extend between a first end **143** and a second end **144** along the longitudinal axis **141**. The deformable portion **145** may be located proximate the first end **143** of the sleeve **142** and the rigid portion **146** may be located proximate the second end **144** of the sleeve **142**. The collar **150** may be configured to slide onto the deformable portion **145** (e.g., from the first end **143**) and move the deformable portion **145** to compress the sleeve **142** onto the substrate **116** such that movement of the substrate **116** relative to the sleeve **142** is restricted. In other words, the collar **150** may apply a compressing force on the deformable portion **145** of the sleeve **142** such that the deformable portion may apply a compressing force on the substrate **116**.

The deformable portion **145** of the sleeve **142** may be deformable in any suitable way. For example, as shown in FIGS. **3** and **4A**, the sleeve **142** may define one or more slots **148** through the deformable portion **145**. The one or more slots **148** may extend through the sleeve **142** along a plane in which the longitudinal axis **141** lies. The one or more slots **148** define an absence of material of the sleeve **142** such that portions of the sleeve **142** on either side of the one or more slots **148** may compress towards one another (e.g., towards the gap defined by the one or more slots **148**) when a force is applied inward on the outer surface. Therefore, the portions of the sleeve **142** on opposite sides of the one or more slots **148** may move to compress or squeeze on the substrate **116** when the collar **150** is slid over the sleeve **142** (e.g., the deformable portion **145**).

The one or more slots **148** may extend from the first end **143** of the sleeve **142** and towards the rigid portion **146** along the longitudinal axis **141**. The one or more slots **148** may extend for the entire length of the deformable portion **145** of the sleeve **142** or for a length less than the entire deformable portion **145** (e.g., along the longitudinal axis **141**). The one or more slots **148** may not extend within the rigid portion **146** of the sleeve **142**.

Additionally, in one or more embodiments, the sleeve **142** (e.g., an outer surface of the sleeve **142**) may define an increasing taper from an end (e.g., the first end **143**) of the sleeve **142** as shown in FIG. **3**. The taper may define any suitable angle to the longitudinal axis **141**. In one or more

embodiments, the outer surface of the sleeve **142** may define no taper and extend parallel to the longitudinal axis **141**. The inner hole **152** of the collar **150** may define a substantially fixed diameter such that when the collar **150** is slid over the sleeve **142** from the end (e.g., the first end **143**), the collar **150** may compress the sleeve **142** as the collar **150** moves further onto the sleeve **142** (e.g., as the collar **150** progresses over the increased taper of the sleeve **142**). In other words, as the collar **150** moves along an increased diameter (e.g., because of the taper) of the sleeve **142**, the collar **150** may apply a force to the outer surface of the sleeve **142** (e.g., the deformable portion **145**) and compress the deformable portion **145** of the sleeve **142** (e.g., because of the one or more slots **148**). The compression of the sleeve **142** may then be transferred to compression of the substrate **116** passing through the sleeve **142** to maintain the substrate **116** in place.

The collar **150** may be positioned on the sleeve **142** in any suitable way. For example, the collar **150** may be slid onto the sleeve **142** manually or through the use of a motor, as described herein. The substrate **116** may be inserted into the opening **149** of the sleeve **142** and the collar **150** may be slid onto the sleeve **142** to engage the clamp **140** and restrict movement of the substrate **116** (e.g., relative to the clamp **140**). Also, the collar **150** may be slid off of the sleeve **142** to disengage the clamp **140** and allow movement of the substrate **116** (e.g., relative to the clamp **140**).

The clamp **140** (e.g., the sleeve **142**) may define a plurality of openings **149** configured to receive a plurality of substrates **116**, as shown in FIG. 4A. In other words, the sleeve **142** may include any number of openings **149** to correspond to the desired number of substrates **116**. For example, as shown in FIG. 4A, the sleeve **142** defines five openings **149** to receive five substrates **116**. Therefore, in such embodiments, melted filament material may be printed on the five substrates to form a device having five lumens (e.g., corresponding to the substrates **116**). The openings **149** of the sleeve **142** may assist in maintaining spatial orientation of the plurality of substrates **116** to form a more consistent resulting catheter or device.

The sleeve **142** may define any suitable number of openings **149**. Further, the one or more slots **148** may align with the openings **149** (e.g., extending through a center of the openings **149**) so that the sleeve **142** may be compressed to restrict movement of the substrate **116** extending through the corresponding opening **149**. In other words, each opening **149** may have two separate portions of the sleeve **142** that move relative to one another and compress or squeeze a substrate **116** in the opening **149**. Further, the one or more slots **148** may pass through multiple openings **149**. Therefore, each opening **149** may include at least one slot **148** extending therethrough to assist in clamping the substrates **116**.

Further, the clamp **140** (e.g., the sleeve **142** and/or the collar **150**) may be operably coupled to a substrate motor. In some embodiments, the substrate motor may be used to control engaging and disengaging of the clamp **140**. Referring back to FIG. 1, in some embodiments, the substrate motor may be used to rotate the substrate **116** in a clockwise or counterclockwise direction about a longitudinal axis **126**. For example, a translation motor may be operably coupled between a stock **120**, **122** and the platform **124**. In some embodiments, the translation motor may be used to translate the stock **120**, **122** in a longitudinal direction along the longitudinal axis **126**. In some embodiments, the translation motor also may be used to translate the stock **120**, **122** in a lateral direction different than the longitudinal axis **126**. The

lateral direction may be oriented substantially orthogonal, or perpendicular, to the longitudinal axis **126**.

In some embodiments, the substrate handling system **108** may be configured to move the head stock **120** at least in a longitudinal direction (for example, parallel to the longitudinal axis **126**) relative to the platform **124**. The substrate **116** may be fed through the substrate channel of the heating cartridge **102** by movement of the head stock **120** relative to the platform **124**. A distal portion of the substrate **116** may be clamped into the head stock **120** as described herein. The head stock **120** may be positioned close to the heating cartridge **102** at the beginning of the jacket formation process. The head stock **120** may move distally away from the heating cartridge **102**, for example, in a direction parallel to the longitudinal axis **126**. In other words, the head stock **120** may move toward the distal region **128** of the system **100** while pulling the secured substrate **116** through the heating cartridge **102**. As the substrate **116** passes through the heating cartridge **102**, melted filament material from the filament **114** may be formed or deposited onto the substrate **116** to form the jacket **118**. The heating cartridge **102** may be stationary relative to the platform **124**. In some embodiments, the tail stock **122** may be omitted.

In some embodiments, the substrate handling system **108** may be configured to move the heating cartridge **102** at least in a longitudinal direction (along the longitudinal axis **126**) relative to the platform **124**. The substrate **116** may be fed through the substrate channel of the heating cartridge **102**. A distal portion of the substrate **116** may be clamped into the head stock **120** as described herein. Further, in one or more embodiments, a proximal portion of the substrate **116** may be clamped into the tail stock **122**. In one example, the heating cartridge **102** may be positioned relatively close to the head stock **120** at the beginning of the jacket formation process. The heating cartridge **102** may move proximally away from the head stock **120**. The heating cartridge **102** may move toward the proximal region **130** of the system **100**. As the heating cartridge **102** passes over the substrate **116**, melted filament material may be deposited onto the substrate **116** to form a jacket **118**. The head stock **120** and the tail stock **122** may be stationary relative to the platform **124**. In another example, the heating cartridge **102** may start near the tail stock **122** and move toward the distal region **128**.

In one or more embodiments, the heating cartridge **102** may make multiple passes along the substrate **116** to form more than one jacket or layer. For example, once the heating cartridge **102** moves along the length of the substrate **116** to form a first jacket, the heating cartridge **102** may move along the length of the substrate **116** again and begin forming a second jacket (e.g., forming an inner layer and a middle/outer layer or jacket). Further, before forming the second jacket, a variety of different components may be positioned on the first jacket. Further yet, in one or more embodiments, the heating cartridge **102** may make more than two passes to form any suitable number of layers or jackets. Therefore, the resulting device (e.g., formed by multiple layers or jackets) may allow for changes along the component radially or a non-symmetrical application relative to the cross-section.

One or more motors of the substrate handling system **108** may be used to rotate one or both of the substrate **116** and the heating cartridge **102** relative to one another. In some embodiments, only the substrate **116** may be rotated about the longitudinal axis **126**. In some embodiments, only the heating cartridge **102** may be rotated about the longitudinal

axis **126**. In some embodiments, both the substrate **116** and the heating cartridge **102** may be rotated about the longitudinal axis **126**.

The heating cartridge **102** may be part of an assembly **132**. The assembly **132** may be coupled to the platform **124**. In some embodiments, one or more motors of the substrate handling system **108** may be coupled between assembly **132** and the platform **124** to move (e.g., translate or rotate) the assembly **132**, including the heating cartridge **102**, relative to the platform **124** or the substrate **116**. In some embodiments, one or more motors of the substrate handling system **108** may be coupled between a frame of the assembly **132** and the heating cartridge **102** to move (e.g., translate or rotate) the heating cartridge relative to the platform **124**.

In some embodiments, the substrate **116** may be rotated about the longitudinal axis **126** relative to the heating cartridge **102** to facilitate forming certain structures of the jacket **118**. In one example, the substrate **116** may be rotated by one or both of the head stock **120** and the tail stock **122** of the substrate handling system **108**. In another example, the heating cartridge **102** or subassembly **132** may be rotated by the substrate handling system **108**.

The system **100** may include one or more concentricity guides **134**. The concentricity guide **134** may facilitate adjustments to the concentricity of the jacket **118** around the substrate **116** before or after the substrate passes through the heating cartridge **102**. The concentricity guide **134** may be longitudinally spaced from the heating cartridge **102**. In some embodiments, the spacing may be greater than or equal to 1, 2, 3, 4, or 5 cm. The spacing may be sufficient to allow the jacket **118** to cool down and no longer be deformable. In some embodiments, one or more concentricity guides **134** may be positioned proximal to the heating cartridge **102** to engage the substrate **116**. The concentricity guide **134** may mitigate drooping of the substrate **116** and may mitigate susceptibility to eccentricity in the alignment of the stock **120**, **122** and the heating cartridge **102**.

Any suitable implementation may be used to provide the filament handling system **106**. One or more filaments **114** may be loaded into the filament handling system **106**. For example, filaments **114** may be provided in the form of wound coils. Filaments **114** may be fed to the heating cartridge **102** by the filament handling system **106**. In some embodiments, the filament handling system **106** may include one, two, or more pinch rollers to engage the one or more filaments **114**. In some embodiments, the filament handling system **106** may include one or more motors. The one or more motors may be coupled to the one or more pinch rollers to control rotation of the pinch rollers. The force exerted by the motors onto the pinch rollers and thus onto the one or more filaments **114** may be controlled by the controller **110**.

In some embodiments, the filament handling system **106** may be configured to feed the filaments **114** including at least a first filament and a second filament. The jacket **118** may be formed from the material of one or both of the filaments **114**. The filament handling system **106** may be capable of selectively feeding the first filament and the second filament. For example, one motor may feed the first filament and another motor may feed the second filament. Each of the motors may be independently controlled by the controller **110**. Selective, or independent, control of the feeds may allow for the same or different feed forces to be applied to each of the filaments **114**.

The filaments **114** may be made of any suitable material, such as polyethylene, PEBAX elastomer (commercially available from Arkema S.A. of Colombes, France), nylon 12, polyurethane, polyester, liquid silicone rubber (LSR), or PTFE.

The filaments **114** may have any suitable Shore durometer. In some embodiments, the filaments **114** may have, or define, a Shore durometer suitable for use in a catheter. In some embodiments, the filaments **114** have a Shore durometer of at least 25A and up to 90A. In some embodiments, the filaments **114** have a Shore durometer of at least 25D and up to 80D.

In some embodiments, the filament handling system **106** may provide a soft filament as one of the filaments **114**. In some embodiments, a soft filament may have a Shore durometer less than or equal to 90A, 80A, 70A, 80D, 72D, 70D, 60D, 50D, 40D, or 35D.

In some embodiments, the filament handling system **106** may provide a hard filament and a soft filament having a Shore durometer less than the soft filament. In some embodiments, the soft filament has a Shore durometer that is 10D, 20D, 30D, 35D, or 40D less than a Shore durometer of the hard filament.

The system **100** may be configured to provide a jacket **118** between the Shore durometers of a hard filament and a soft filament. In some embodiments, the filament handling system **106** may provide a hard filament having a Shore durometer equal to 72D and a soft filament having a Shore durometer equal to 35D. The system **100** may be capable of providing a jacket **118** having a Shore durometer that is equal to or greater than 35D and less than or equal to 72D.

The system **100** may be configured to provide a jacket **118** having, or defining, segments with different Shore durometers. In some embodiments, the system **100** may be capable of providing a jacket **118** having one or more of a 35D segment, a 40D segment, 55D segment, and a 72D segment.

The filaments **114** may have any suitable width or diameter. In some embodiments, the filaments **114** have a width or diameter of 1.75 mm. In some embodiments, the filaments **114** have a width or diameter of less than or equal to 1.75, 1.5, 1.25, 1, 0.75, or 0.5 mm.

Segments may have uniform or non-uniform Shore durometers. The system **100** may be configured to provide jacket **118** having one or more segments with non-uniform Shore durometers. In some embodiments, the jacket **118** may include continuous transitions between at least two different Shore durometers.

The controller **110** may be configured to change a feeding force applied to one or more of the filaments **114** to change a ratio of material in the jacket over a longitudinal distance. By varying the feeding force, the system **100** may provide different Shore durometer segments in a jacket **118**, whether uniform or non-uniform. In one example, sharp transitions between uniform segments may be provided by stopping or slowing longitudinal movement while continuously, or discretely with a large step, changing the feeding force of one filament relative to another filament of the substrate **116** relative to the heating cartridge **102**. In another example, gradual transitions between segments may be provided by continuously, or discretely with small steps, changing the feeding force of one filament relative to another filament while longitudinally moving the substrate **116** relative to the heating cartridge **102**.

The system **100** may be configured to provide a jacket **118** having varying thicknesses. In some embodiments, the controller **110** may be configured to vary one or more parameters, for example, at least one of: a longitudinal speed of the

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substrate **116** relative to the heating cartridge **102**, a feeding force applied to one or more filaments **114**, and an amount of heat provided by the heating element **104**. Varying one or more of these parameters during formation of the jacket **118** may be used to change a thickness of the jacket over a longitudinal distance. In some embodiments, the controller **110** may be configured to vary one or more of these parameters in conjunction with using a particular heating cartridge.

The one or more wires **115** provided by the wire handling system **107** may be introduced in any suitable manner. In some embodiments, the wires **115** may be attached to the substrate **116** and pulled by movement of the substrate. One example of a wire is a pull wire that may be used to steer the catheter produced by the system **100**. In some embodiments, a particularly shaped heating cartridge may be used to accommodate one or more wires **115**.

Any suitable type of heating element **104** may be used. In some embodiments, the heating element **104** may be a resistive-type heating element, which may provide heat in response to an electrical current. Other types of heating elements that may be used for the heating element **104** include a radio frequency (RF) or ultrasonic-type heating element. The heating element **104** may be capable of providing heat sufficient to melt the filaments **114**. In some embodiments, the heating element **104** may heat the filaments **114** to greater than or equal to 235, 240, 250, or 260 degrees Celsius. In general, the one or more heating elements **104** may be used to heat the filaments **114** to any suitable melting temperature known to one of ordinary skill in the art having the benefit of this disclosure.

Any suitable user interface **112** may be used to communicate with the controller **110**. Non-limiting examples of user interfaces **112** include one or more of a stationary or portable computer, a monitor or other display, a touchscreen, a keyboard, a mouse, a tablet, a phone, a knob, a switch, a button, and the like. In some embodiments, the user interface **112** may allow the user to input direct commands to or to enter code to program operations of the controller **110**. For example, in one or more embodiments, a program may include various command lines (e.g., for printing various jacket segments) as described in U.S. patent application Ser. No. 17/081,815, entitled "ADDITIVE MANUFACTURING FOR MEDICAL DEVICES," which is herein incorporated by reference in its entirety.

FIG. 2 shows one example of an additive manufacturing apparatus **200** of the additive manufacturing system **100** in an end view along the longitudinal axis **126**, which is illustrated as a circle and cross. More detail of some components of the additive manufacturing system **100** are shown, such as the heating cartridge **102** and the filament handling system **106**.

The heating cartridge **102** may include a heating block **202** at least partially defining an interior volume **204**. The interior volume **204** may be heated by the heating element **104**. The heating element **104** may be thermally coupled to the heating block **202** to melt filament material in the interior volume **204**. In general, the system **100** may be configured to melt any portion of the filaments **114** in the interior volume **204**. The heating element **104** may be disposed in an exposed or exterior volume defined in the heating block **202**. The heating element **104** may be positioned proximate to or adjacent to the interior volume **204**. In some embodiments, one, two, three, or more heating elements **104** may be thermally coupled to the heating block **202**.

The heating block **202** may allow the substrate **116**, which may be an elongate substrate or member, to pass through the

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heating block. The substrate **116** may be able to extend, or pass, through the interior volume **204**. The substrate channel **206** defined by the heating cartridge **102** may extend through the interior volume **204**. The substrate channel **206** may extend in a same or similar direction as the substrate **116**. The substrate channel **206** may extend along the longitudinal axis **126**.

A width or diameter of the interior volume **204** is larger than a width or diameter of the substrate **116**. The width or diameter of the interior volume **204** or the substrate **116** is defined in a lateral direction, which may be orthogonal to the longitudinal axis **126**. In one example, the lateral direction may be defined along a lateral axis **210**. In some embodiments, the clearance between the substrate **116** and interior volume **204** is relatively small to facilitate changes in composition of filament material used to form the jacket **118** (FIG. 1) around the substrate **116**.

The portion of the interior volume **204** around the substrate **116** may receive a flow of melted filament material from the filaments **114**. When more than one filament material is provided to the interior volume **204**, the filament materials may flow and blend, or mix, around the substrate **116**.

In the illustrated embodiment, the filaments **114** includes a first filament **212** and a second filament **214**. The first filament **212** may be provided into the interior volume **204** through a first filament port **216** at least partially defined by the heating block **202**. The second filament **214** may be provided into the interior volume **204** through a second filament port **218** at least partially defined by the heating block **202**. Each filament port **216**, **218** may be at least partially defined by the heating block **202**. Each filament port **216**, **218** may be in fluid communication with the interior volume **204**.

The filaments **114** may be delivered to the interior volume **204** in the same or different manners. In the illustrated embodiment, the first filament **212** is delivered to the interior volume **204** in a different manner than the second filament **214**.

The filament handling system **106** may include a first handling subassembly **220**. The first handling subassembly **220** may deliver the first filament **212** to the interior volume **204**. The first handling subassembly **220** may include one or more pinch rollers **222**. Each of the one or more pinch rollers **222** may be operably coupled to a motor. Any suitable number of pinch rollers **222** may be used. As illustrated, the first handling subassembly **220** may include two sets of pinch rollers **222**. Pinch rollers **222** may be used to apply a motive force to the first filament **212** to move the first filament, for example, toward the interior volume **204**.

The heating cartridge **102** may include a first guide sheath **224**. The first guide sheath **224** may extend between the filament handling system **106** and the interior volume **204**. The first guide sheath **224** may be coupled to the heating block **202**. The first guide sheath **224** may extend into the first filament port **216** from an exterior of the heating block **202**. The first guide sheath **224** may define a lumen in fluid communication with the interior volume **204**. An inner width or diameter of the lumen may be defined to be greater than a width or diameter of the first filament **212**. The first filament **212** may extend through the first guide sheath **224** from the pinch rollers **222** of the first handling subassembly **220** to the first filament port **216** and extend distally past the first guide sheath **224** into the interior volume **204**.

As used herein with respect to the filaments **114**, the term "distal" refers to a direction closer to the interior volume **204**

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and the term “proximal” refers to a direction closer to the filament handling system 106.

In some embodiments, a proximal end of the first guide sheath 224 may terminate proximate to one of the pinch rollers 222. A distal end of the first guide sheath 224 may terminate at a shoulder 226 defined by the first filament port 216. A distal portion or distal end of the first guide sheath 224 may be positioned proximate to or adjacent to the interior volume 204.

The inner width or diameter of the lumen of the first guide sheath 224 may be defined to be substantially the same or equal to an inner width or diameter of the first filament port 216, such as a smallest inner width or diameter of the first filament port. In other words, an inner surface of the first guide sheath 224 may be flush with an inner surface of the first filament port 216.

In some embodiments, the heating cartridge 102 may include a support element 228. The support element 228 may be coupled to the first guide sheath 224. The first guide sheath 224 may extend through a lumen defined by the support element 228. The support element 228 may be proximate to the heating block 202. In the illustrated embodiment, the support element 228 is coupled to the heating block 202. The support element 228 may include a coupling protrusion configured to be mechanically coupled to a coupling receptacle 230 defined by the first filament port 216. In some embodiments, the coupling receptacle 230 may define threads and the coupling protrusion of the support element 228 may define complementary threads.

The coupling receptacle 230 may terminate at the shoulder 226 of the first filament port 216. The coupling protrusion of the support element 228 may be designed to terminate at the shoulder 226. In some embodiments, a distal end of the support element 228 and the distal end of the first guide sheath 224 may engage the shoulder 226. In other embodiments, the distal end of the support element 228 may engage the shoulder 226 and the distal end of the first guide sheath 224 may engage a second shoulder (not shown) defined by the first filament port 216 distal to the shoulder 226.

When the first filament port 216 defines one shoulder, the first filament port 216 may define at least two different inner widths or diameters. The larger inner width or diameter may be sized to thread the support element 228 and the smaller inner width or diameter may be sized to match the inner width or diameter of the first guide sheath 224.

When the second filament port 218 defines two shoulders, the first filament port 216 may define at least three different inner widths or diameters. The largest inner width or diameter may be sized to thread the support element 228. The intermediate inner width or diameter may be sized to receive a distal portion of the first guide sheath 224. The smallest inner width or diameter may be sized to match the inner width or diameter of the first guide sheath 224.

The filament handling system 106 may include a second handling subassembly 232. The second handling subassembly 232 may deliver the second filament 214 to the interior volume 204. The second handling subassembly 232 may include one or more pinch rollers 222. Each of the one or more pinch rollers 222 may be operably coupled to a motor. Any suitable number of pinch rollers 222 may be used. As illustrated, the second handling subassembly 232 may include one set of pinch rollers 222. Pinch rollers 222 may be used to apply a motive force to the second filament 214.

The heating cartridge 102 may include one or more of a second guide sheath 234, a heat sink 236, and a heat break 238. The second guide sheath 234 may extend at least

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between the second handling subassembly 232 and the heat sink 236. The second guide sheath 234 may be coupled to the heat sink. The second guide sheath 234 may be coupled to the second handling subassembly 232. The heat sink 236 may be coupled to the heat break 238. The heat break 238 may be coupled to the heat block 202. The heat break 238 may extend into the second filament port 218 from an exterior of the heating block 202.

The second guide sheath 234 may define a lumen in fluid communication with the interior volume 204. The second filament 214 may extend through the second guide sheath 234 from the second handling subassembly 232 to the heat sink 236, through the heat sink 236, through the heat break, and through the second filament port 218. In some embodiments, the second guide sheath 234 may extend to the pinch rollers 22 in the second handling subassembly 232. In some embodiments, the second guide sheath 234 may extend at least partially into the heat sink 236.

The heat break 238 may be proximate to the heating block 202. The heat break 238 may be positioned between the heat sink 236 and the heating block 202. The heat break 238 may include a coupling protrusion configured to mechanically couple to a coupling receptacle 240 defined by the second filament port 218. In some embodiments, the coupling receptacle 240 may define threads and the coupling protrusion of the heat break 238 may define complementary threads. The second filament port 218 may include one or more shoulders such as those described with respect to the first filament port 216, except that the second filament port 218 may not be configured to receive the second guide sheath 234. The inner width or diameter of the support element 228 may be larger than the inner width or diameter of the heat break 238, for example, to accommodate the outer width or diameter of the first guide sheath 224. In other embodiments, the second filament port 218 may be configured to receive the second guide sheath 234 in a similar manner to the first filament port 216 receiving the first guide sheath 224.

Any suitable material may be used to make the guide sheaths 224, 234. In some embodiments, one or both guide sheaths 224, 234 may include a synthetic fluoropolymer. One or both guide sheaths 224, 234 may include polytetrafluoroethylene (PTFE). Another suitable material may include an ultra-high molecular weight polyethylene (UHMWPE).

Any suitable material may be used to make the support element 228. In some embodiments, the support element 228 may be a thermal insulator. The support element 228 may include a thermoplastic. The support element 228 may be made of a polyamide-imide, such as a TORLON polyamide-imide (commercially available from McMaster-Carr Supply Co. of Elmhurst, Illinois). Other suitable materials may include liquid-crystal polymer, polyaryletherketone (PAEK), polyphenylene sulfide, and polysulfone.

The support element 228 may provide mechanical support to the first guide sheath 224. The support element 228 may include a substantially rigid material. In some embodiments, the support element 228 include a material having a higher durometer than material used to make the first guide sheath 224.

Any suitable material may be used to make the heat sink 236. The heat sink 236 may include a high thermal conductivity material. In some embodiments, the heat sink 236 includes aluminum.

Any suitable material may be used to make the heat break 238. The heat break 238 may include a low thermal conductivity material. In some embodiments, the heat break 238

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includes titanium. The heat break **238** may include a necked portion to reduce the amount of material between a proximal portion and a distal portion of the heat break. The necked portion may facilitate a reduced thermal conductivity between the proximal portion and the distal portion of the heat break **238**.

In general, use of the apparatus **200** may facilitate using softer filaments at high feed forces and pressures, which tend to compress the soft filament and may result in jamming. Using higher feed forces and pressures may allow for a greater range of process conditions and may provide a consistent jacket around the substrate. In particular, use of the first guide sheath **224** extending at least partially into the first filament port **216** may facilitate the use of softer filament and greater “push-ability.” Additionally, or alternatively, the use of the support element **228** may also facilitate the use of softer filament and greater “push-ability.”

FIG. **5** shows a partial cross-sectional side view of one example of the heating cartridge **102**. The heating cartridge **102** or the heating block **202** may extend from a proximal side **410** to a distal side **412**. In some embodiments, the heating cartridge **102** may include one or more of the heating block **202**, an inlet die **402** coupled to the proximal side **410** of the heating block, an outlet die **404** coupled to the distal side **412** of the heating block, a proximal retaining plate **406** to facilitate retaining the inlet die adjacent to the heating block, and a distal retaining plate **408** to facilitate retaining the outlet die adjacent to the heating block.

The inlet die **402** and the outlet die **404** may be retained in any suitable manner. In the illustrated embodiment, the outlet die **404** may be retained by a distal shoulder of the distal retaining plate **408**. In some embodiments, the inlet die **402** may be retained by the proximal retaining plate **406** between a distal shoulder of the proximal retaining plate **406** and a fastener, such as a nut with a lumen extending through, which may be threaded to the retaining plate to engage a proximal surface of the inlet die. The retaining plates **406**, **408** may be fastened to the heating block **202** in any suitable manner.

The inlet die **402** may at least partially define a substrate inlet port **414**. The outlet die **404** may at least partially define a substrate outlet port **416**.

The inlet die **402** may at least partially define the interior volume **204**. The outlet die **404** may at least partially define the interior volume **204**. In some embodiments, an exterior surface of the inlet die **402**, an interior surface of the outlet die **404**, and an interior surface of the heating block **202** may cooperatively define the interior volume **204**.

The substrate channel **206** may be described as extending from the proximal side **410** to the distal side **412** of the heating cartridge **102**, or vice versa. The substrate channel **206** may extend through the interior volume **204**. As shown, the substrate channel **206** may extend through one or more of the proximal retaining plate **406**, the inlet die **402**, the heating block **202**, the outlet die **404**, and the distal retaining plate **408**.

FIG. **6** illustrates an end view of one example of an inlet die **402** that may be used in the heating cartridge **102**. The inlet die **402** may include a plurality of openings **418** to receive the substrates **116**. In other words, the substrates **116** may extend through the inlet die **402** to position the substrates and maintain spatial orientation of the plurality of substrates **116**. Therefore, the substrates **116** may be properly positioned within the interior volume **204** for filament material to be printed thereon. The openings **418** of the inlet die **402** may match any suitable the number of substrates **116**. As shown in FIG. **6**, the inlet die **402** defines five

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openings **418**. Further, the openings **418** of the inlet die **402** may match the openings **149** of the sleeve **142** such that, e.g., the substrates **116** may pass through and extend therebetween.

FIG. **7** illustrates a subassembly **160** configured to be positioned over the substrate **116** (e.g., not shown in FIG. **7**) and printed thereupon by the additive manufacturing system **100** described herein. In one or more embodiments, the subassembly **160** may be positioned on the substrate **116** with an insulation layer or liner therebetween. The subassembly **160** may include a variety of different components that can be embedded in the medical device by printing a jacket **118** (e.g., a catheter jacket or outer insulation) over the subassembly **160**. For example, the subassembly **160** may include electrical components or circuits, braided or coiled shafts, polytetrafluoroethylene (PTFE), fluoroethylene propylene (FEP), polymer tubing, pull-wire assemblies, polyurethane, carbothane, sensors, inner lumens, etc. Specifically, in one or more embodiments, electrical components of the subassembly **160** (e.g., for a lead body) may include electrode rings, coils, wires, cables, sensors, electromagnetic coils and/or cables, fiber optic cables, circuits, insulation layers, adhesives, joints between components, etc.

As shown in FIG. **7**, the subassembly **160** may extend between a first end **161** and a second end **162** along a longitudinal axis **169**. Further, the subassembly **160** may define a passageway **165** extending between the first and second ends **161**, **162** and configured to receive the substrate **116** (not shown in FIG. **7**). In other words, the subassembly **160** may be positioned on and encircle the substrate **116**. The subassembly **160** may then be printed over the subassembly **160** to encapsulate or cover at least a portion of the subassembly **160**. In one or more embodiments, the medical device may include multiple subassemblies **160** contained therein and, e.g., encapsulated by the printed jacket **118**.

The subassembly **160** may include an electrode ring **164** proximate the first end **161** of the subassembly **160**. When the system **100** prints over the substrate **116** and subassembly **160** to create the medical device, the electrode ring **164** may be left exposed. For example, in one or more embodiments, the system **100** may selectively print along the subassembly **160** such that the jacket **118** is not printed over the electrode ring **164** (e.g., to leave the electrode ring **164** exposed). In one or more embodiments, a mask (e.g., an etchable mask) may be deposited over the electrode ring **164** such that after the jacket **118** is printed over the subassembly **160**, the portion of the printed jacket **118** positioned over the electrode ring **164** may be removed or etched away (e.g., to leave the electrode ring **164** exposed). The electrode ring **164** may be left exposed such that the medical device may deliver therapy at the location at which the electrode ring **164** is positioned. Further, the subassembly **160** may include any number of electrode rings **164**. Further yet, the electrode rings **164** may be positioned at any suitable location along the subassembly **160**.

The subassembly **160** may also include one or more coils to transmit an electrical signal through the device and the subassembly **160**. For example, as shown in FIG. **7**, the subassembly may include an inner coil **166** and an outer coil **168**. Additionally, in some embodiments, the subassembly **160** may include insulation, cables, wires, pull-wires, sensors, lumens (e.g., open lumens for fluid transfer or shaping members), adhesives, etc. Further, as shown in FIG. **7**, at least a portion of the printed jacket **118** is positioned over the inner coils **166**.

FIG. **8** shows one example of a method **300** of using the system for additive manufacturing to print a jacket over a



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substrate and an electrical subassembly positioned thereon. The method may include positioning **310** an electrical subassembly on a substrate.

For example, FIG. 9A illustrates a conceptual diagram of a subassembly **160** positioned on a substrate **116** (e.g., a conceptual cross-section of a tubular subassembly **160** and substrate **116**). The subassembly **160** may define a pre-formed body upon which various components are located. The body of the subassembly **160** may include (e.g., be formed of) metals and alloys, thermoplastics (such as polyurethane, silicone, carbothane, conductors, insulators, electrodes, etc. The subassembly **160** may include an electrode ring **164** proximate a first end **161** of the subassembly **160**. The subassembly **160** may also include an inner coil **166**, an outer coil **168**, and insulation **167** therebetween. The insulation **167** may electrically separate and isolate the inner and outer coils **166**, **168** from one another. For example, the insulation **167** may include (e.g., be formed of) polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene copolymer (ETFE), silicone, polyurethane, adhesive layers, polymer tubing, or other non-conductive materials. One or both of the inner and outer coils **166**, **168** may be operably connected to the electrode ring **164** or other electrode rings of the subassembly **160**. It is noted that, while only one electrode ring **164** is shown in FIG. 9A, the subassembly **160** may include any suitable number of electrode rings. Further, the inner and outer coils **166**, **168** may be operably connected to other components of the medical device (e.g., a lead extending through the printed jacket) to provide a connection between an electrical source and the electrode rings **164**.

The subassembly **160** may also include a bonding zone **170** located adjacent to the electrode ring **164**. The bonding zone **170** may be located along the subassembly **160** between the electrode ring **164** and the inner/outer coils **166**, **168** (e.g., along the longitudinal axis **169**). The jacket may be printed to cover at least a portion of the bonding zone **170** and extend over the inner and outer coils **166**, **168**. Further, the jacket may be printed on the bonding zone **170** (and away from the electrode ring **164**) such that the printed jacket may be spaced apart from the electrode ring **164**. The bonding zone **170** may include (e.g., be formed of) platinum-iridium alloy, titanium, MP35N, stainless steel, copper, welds, chemical joints (e.g., such as adhesive joints, thermal joints, etc.), mechanical interlock, etc.

The method **300** of FIG. 8 may also include feeding **320** the substrate through a substrate channel in a heating cartridge. The substrate channel may be in fluid communication with an interior cavity of the heating cartridge. For example, as shown in FIG. 9B, the substrate **116** and the subassembly **160** positioned thereon may be located proximate the heating cartridge **102** to be passed therethrough. In one or more embodiments, the subassembly **160** may include a mask **172** surrounding the electrode ring **164** (e.g., as shown in FIG. 9B). The mask **172** may be positioned over the electrode ring **164** to protect the electrode ring **164** during the printing process (e.g., prevent material from being directly printed on the electrode ring **164**). Further, the mask **172** may be removed (e.g., etched) after the printing process to remove anything printed thereon and expose the electrode ring **164** (e.g., to be used as a contact point for the resulting medical device). The mask **172** may include (e.g., be formed of) any suitable material such as, e.g., fluorinated ethylene propylene (FEP), polytetrafluoroethylene (PTFE), other thermoplastics, polyurethane, polyether block amid (PEBA), ethylene tetrafluoroethylene copolymer (ETFE), PET heat shrink, etc. As such, in one or more embodiments, the

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method **300** of FIG. 8 may also include masking the electrode ring of the electrical subassembly. In other embodiments, the system may selectively print to avoid the electrode ring (e.g., by starting printing on the bonding zone **170** or away from the electrode ring **164**).

Additionally, the method **300** may include feeding **330** at least a first filament through a filament port into the interior cavity and melting **340** the first filament in the interior cavity. In one or more embodiments, the method may also include feeding a second filament through another filament port into the interior cavity and melting the second filament with the first filament to form the printed jacket including material from at least the first filament and the second filament. In other words, multiple filaments may be used in combination to achieve various different properties and characteristics of the resulting printed jacket, as described herein. For example, in one or more embodiments, the method may also include adjusting a ratio of the first filament relative to the second filament over a longitudinal distance (e.g., of the printed jacket) to change the hardness (e.g., Shore durometer) of the printed jacket over the longitudinal distance of the printed jacket.

The method **300** of FIG. 8 may also include moving **350** the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket including material from at least the first filament. For example, as shown in FIG. 9C, the subassembly **160** and substrate **116** may pass through the heating cartridge **102** to print a material over the subassembly **160** and substrate **116** to form a printed jacket **118**. The printed jacket **118** may cover at least a portion of the (electrical) subassembly **160** and the substrate **116** to help create the medical device. Specifically, as shown in FIG. 9C, the jacket **118** may be printed over a mask **172** covering the electrode ring **164**. Also, in one or more embodiments, the heating cartridge **102** or substrate **116** may also be rotated relative to one another while moving along the longitudinal directional.

Further, as shown in FIG. 9D, the mask **172** may be removed or etched to remove the printed jacket **118** from the electrode ring **164**. Thereafter, the electrode ring **164** may be exposed and the printed jacket **118** may be spaced apart from and/or not covering the electrode ring **164**. As such, in one or more embodiments, the method **300** of FIG. 8 may also include removing the mask to expose the electrode ring after the printed jacket is formed (e.g., over the mask). Also, in some embodiments, the jacket **118** may be printed on the subassembly **160** (e.g., without a mask) such that no material is printed on the electrode ring **164** (e.g., starting printing on the bonding zone **170**) and the printed jacket **118** is spaced apart from the electrode ring **164**.

After printing the jacket **118** on the substrate **116** and the subassembly **160**, the substrate **116** may be removed from the printed jacket **118** to define a lumen extending through the printed jacket **118**. In one or more embodiments, an electrical lead may be inserted into the lumen to form the medical device. In one or more embodiments, the substrate **116** being printed on may already be an electrical lead (e.g., such that the substrate **116** may not need to be removed and replaced after printing). In other embodiments, the substrate may be replaced with (e.g., within the lumen), e.g., a guidewire, stylet, shaping member, radiopacifying member, fluid, a pull wire, coil, cable, malleable metal or plastic, a sensor (e.g., a fiberoptic cable), thermocouple assembly, etc.

FIG. 10 illustrates one example of a jacket **118** printed on a substrate **116** that may be manufactured using the system **100** as described herein. The substrate **116** may include a liner **180** (e.g., a harvestable liner) positioned over the

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substrate **116**. The liner **180** may extend around the circumference of the substrate **116**. The substrate **116** may be configured to move relative to the liner **180**. In other words, the substrate **116** may be flexible within the liner **180** even after the jacket **118** is printed over the liner **180**. The liner **180** may include (e.g., be formed of) polytetrafluoroethylene (PTFE), high-density polyethylene (HDPE), polyvinylidene fluoride (PVDF), ethylene tetrafluoroethylene copolymer (ETFE), etc.

The liner **180** may be positioned on the substrate **116** in any suitable way. For example, in one or more embodiments, the liner **180** may be a substance that is deposited on an outer surface of the substrate **116**. In other embodiments, the liner **180** may be a sheath that is slid over the substrate **116**.

The system **100** may be used to print a jacket **118** on the liner **180**. The filament material melted within the heat cartridge may bond with the liner **180** to form the printed jacket **118**. After the printed jacket **118** is formed on the liner **180**, the substrate **116** may still move and be flexible within the liner **180**. Therefore, in one or more embodiments, the substrate **116** may be removed from the printed jacket **118** and a lead, dilator, guidewire, stylet, needle, sensor, conductor, surgical tool, etc. may be inserted therein. Furthermore, in one or more embodiments, the substrate **116** may be an electrical lead. In other words, the jacket **118** may be printed directly on the liner **180** that surrounds the electrical lead. Because the electrical lead maintains movability and flexibility relative to the liner **180** after printing the jacket (e.g., instead of adhering to and limiting movement of the electrical lead), the electrical lead may be used in place of the substrate **116**.

A method **500** of using the system to print a jacket on a liner covered substrate is shown in FIG. **11**. For example, the method **500** may include positioning **510** a liner over a substrate and feeding **520** the liner covered substrate through a substrate channel in a heating cartridge. The substrate channel may be in fluid communication with an interior cavity of the heating cartridge. The method **500** may also include feeding **530** at least a first filament through a filament port into the interior cavity and melting **540** the first filament in the interior cavity. Further, the method **500** may also include moving **550** the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket including material from at least the first filament. The substrate may be configured to move relative to the liner after the printed jacket is formed.

In one or more embodiments, the method may include removing the substrate from the formed printed jacket (e.g., to insert an electrical lead therein).

In one or more embodiments, the method may include feeding a second filament through another filament port into the interior cavity and melting the second filament with the first filament to form the printed jacket including material from at least the first filament and the second filament. Also, in one or more embodiment, the method may include adjusting a ratio of the first filament relative to the second filament over a longitudinal distance to change the hardness (e.g., Shore durometer) of the printed jacket over the longitudinal distance.

#### Illustrative Embodiments

While the present disclosure is not so limited, an appreciation of various aspects of the disclosure will be gained through a discussion of the specific examples and illustrative embodiments provided below. Various modifications of the

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examples and illustrative embodiments, as well as additional embodiments of the disclosure, will become apparent herein.

A1. An additive manufacturing system for producing a medical catheter or lead, the system comprising:

a heating cartridge extending from a proximal side to a distal side and comprising a substrate inlet port at the proximal side and a substrate outlet port at the distal side, the heating cartridge defining an interior volume and a substrate channel extending through the interior volume from the proximal side to the distal side, wherein the heating cartridge defines a first filament port in fluid communication with the interior volume to receive a first filament;

a heating element thermally coupled to the heating cartridge to heat the interior volume;

a filament handling system comprising one or more motors to feed at least the first filament through the first filament port into the interior volume;

a substrate handling system comprising:

a clamp to secure a distal portion of an elongate substrate, wherein the substrate is positioned to pass through the substrate channel when secured by the clamp, wherein the clamp comprises a sleeve and a collar, and wherein the elongate substrate extends through the sleeve and the collar is configured to compress the sleeve on the elongate substrate, and one or more motors to move one or both of the substrate when secured by the clamp and the heating cartridge relative to one another; and

a controller operably coupled to the heating element, one or more motors of the filament handling system, and one or more motors of the substrate handling system, the controller configured to:

activate the heating element to melt any portion of the first filament in the interior volume;

control the one or more motors of the filament handling system to selectively control the feeding of the first filament into the interior volume; and

control one or more motors of the substrate handling system to move one or both of the substrate and the heating cartridge relative to one another in at least a longitudinal direction to form a printed jacket around the substrate, wherein the printed jacket comprises material from the first filament.

A2. The system according to embodiment A1, wherein the heating cartridge comprises an inlet die, a heating block, and an outlet die, wherein the inlet die defines a plurality of openings configured to receive a plurality of substrates, wherein the inlet die maintains spatial orientation of the plurality of substrates.

A3. The system according to any preceding A embodiment, wherein the collar defines an inner hole to receive the sleeve, and wherein the collar is configured to slide along the sleeve to move at least a portion of the sleeve to compress the sleeve against the substrate and restrict movement of the sleeve relative to the substrate.

A4. The system according to any preceding A embodiment, wherein the sleeve comprises a deformable portion and a rigid portion arranged along a longitudinal axis, wherein the collar is configured to slide onto the deformable portion and move the deformable portion to compress the sleeve onto the substrate such that movement of the substrate relative to the sleeve is restricted.

A5. The system according to any preceding A embodiment, wherein the sleeve defines an increasing taper from an end of the sleeve, wherein the collar defines an inner hole having a fixed diameter, wherein the collar is configured to

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slide over the sleeve from the end of the sleeve and compress the sleeve as the collar moves further onto the sleeve.

A6. The system according to any preceding A embodiment, wherein the sleeve defines one or more slots such that at least a portion of the sleeve moves to compress on the substrate when the collar is slid over the sleeve.

A7. The system according to any preceding A embodiment, wherein the sleeve defines a plurality of openings to receive a plurality of substrates, wherein the sleeve maintains spatial orientation of the plurality of substrates.

A8. The system according to any preceding A embodiment, wherein the heating cartridge further defines a first filament port in fluid communication with the interior volume to receive the first filament and a second filament port in fluid communication with the interior volume to receive a second filament, wherein the one or more motors of the filament handling system is configured to feed the second filament through the second filament port into the interior volume, wherein the controller is configured to:

activate the heating element to melt any portion of the second filament in the interior volume, and

control the one or more motors of the filament handling system to selectively control the feeding of the second filament into the interior volume, wherein the printed jacket comprises material from the second filament.

A9. The system according to embodiment A8, wherein the controller is configured to change a feeding force applied to at least one of the first filament and the second filament to change a ratio of material in the printed jacket over a longitudinal distance.

A10. The system according to any preceding A embodiment, wherein the first filament has a Shore durometer less than or equal to 90A, 80A, 70A, 80D, 72D, 70D, 60D, 50D, 40D, or 35D.

A11. The system according to any preceding A embodiment, wherein the substrate outlet port of the heating cartridge defines an interior shoulder, wherein the controller is configured to vary at least one of: a longitudinal speed of the substrate relative to the heating cartridge, a feeding force applied to the first filament, and an amount of heat provided by the heating element to change a thickness of the printed jacket over a longitudinal distance.

B1. A method for additive manufacturing of a medical catheter or lead, the method comprising:

positioning an electrical subassembly on a substrate, wherein the electrical subassembly comprises an electrode ring;

feeding the substrate through a substrate channel in a heating cartridge, the substrate channel in fluid communication with an interior cavity of the heating cartridge;

feeding at least a first filament through a filament port into the interior cavity;

melting the first filament in the interior cavity; and

moving the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket comprising material from at least the first filament, wherein the printed jacket covers at least a portion of the electrical subassembly and is spaced apart from the electrode ring.

B2. The method according to embodiment B1, further comprising:

masking the electrode ring of the electrical subassembly, wherein the printed jacket covers the masked portion of the electrical subassembly; and

removing the mask to expose the electrode ring after the printed jacket is formed.

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B3. The method according to any preceding B embodiment, wherein the electrical subassembly comprises a bonding zone adjacent the electrode ring, and wherein the printed jacket covers at least a portion of the bonding zone.

B4. The method according to embodiment B3, wherein the bonding zone comprises a suitable metal or alloy (e.g., such as platinum-iridium), MP35N, stainless steel, titanium, copper, etc.

B5. The method according to any preceding B embodiment, wherein the electrical subassembly comprises one or more coils (and insulation, wires, cables, polymer tubing, sensors, etc.).

B6. The method according to any preceding B embodiment, further comprising removing the substrate from the printed jacket.

B7. The method according to any preceding B embodiment, further comprising:

feeding a second filament through another filament port into the interior cavity; and

melting the second filament with the first filament to form the printed jacket comprising material from at least the first filament and the second filament.

B8. The method according to embodiment B7, further comprising adjusting a ratio of the first filament relative to the second filament over a longitudinal distance to change the Shore durometer of the printed jacket over a longitudinal distance of the printed jacket.

C1. A method for additive manufacturing of a medical catheter or lead, the method comprising:

positioning a liner over a substrate;

feeding the liner covered substrate through a substrate channel in a heating cartridge, the substrate channel in fluid communication with an interior cavity of the heating cartridge;

feeding at least a first filament through a filament port into the interior cavity;

melting the first filament in the interior cavity; and

moving the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket comprising material from at least the first filament, wherein the substrate is configured to move relative to the liner after the printed jacket is formed.

C2. The method according to embodiment C1, wherein positioning a liner over a substrate comprises depositing a substance on an outer surface of the substrate.

C3. The method according to any preceding C embodiment, wherein positioning a liner over a substrate comprises sliding a sheath over the substrate.

C4. The method according to any preceding C embodiment, wherein the liner comprises a thin-wall fluoroplastic (e.g., such as polytetrafluoroethylene (PTFE), fluoroethylene propylene (FEP), ethylene tetrafluoroethylene copolymer (ETFE), polyvinylidene fluoride (PVDF), high-density polyethylene (HDPE), etc.

C5. The method according to any preceding C embodiment, further comprising removing the substrate from the printed jacket.

C6. The method according to any preceding C embodiment, wherein the substrate comprises a lead.

C7. The method according to any preceding C embodiment, further comprising:

feeding a second filament through another filament port into the interior cavity; and

melting the second filament with the first filament to form the printed jacket comprising material from at least the first filament and the second filament.

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C8. The method according to embodiment C7, further comprising adjusting a ratio of the first filament relative to the second filament over a longitudinal distance to change the Shore durometer of the printed jacket over the longitudinal distance.

D1. A method for additive manufacturing of a medical catheter or lead, the method comprising:

securing a substrate via a clamp, wherein the clamp comprises a sleeve and a collar, and wherein the substrate extends through the sleeve and the collar is configured to compress the sleeve on the substrate;  
feeding the substrate through a substrate channel in a heating cartridge, the substrate channel in fluid communication with an interior cavity of the heating cartridge;  
feeding at least a first filament through a filament port into the interior cavity;  
melting the first filament in the interior cavity; and  
moving the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket comprising material from at least the first filament.

D2. The method according to embodiment D2, wherein the heating cartridge comprises an inlet die, a heating block, and an outlet die, wherein the inlet die defines a plurality of openings configured to receive a plurality of substrates, wherein the inlet die maintains spatial orientation of the plurality of substrates.

D3. The method according to any preceding D embodiment, wherein the collar defines an inner hole to receive the sleeve, and wherein securing the substrate via the clamp comprises sliding the collar along the sleeve to move at least a portion of the sleeve to compress the sleeve against the substrate and restrict movement of the sleeve relative to the substrate.

D4. The method according to any preceding D embodiment, wherein the sleeve comprises a deformable portion and a rigid portion arranged along a longitudinal axis, wherein securing the substrate via the clamp comprises sliding the collar onto the deformable portion and moving the deformable portion to compress the sleeve onto the substrate such that movement of the substrate relative to the sleeve is restricted.

D5. The method according to any preceding D embodiment, wherein the sleeve defines an increasing taper from an end of the sleeve, wherein the collar defines an inner hole having a fixed diameter, wherein securing the substrate via the clamp comprises sliding the collar over the sleeve from the end of the sleeve and compressing the sleeve as the collar moves further onto the sleeve.

D6. The method according to any preceding D embodiment, wherein the sleeve defines one or more slots such that at least a portion of the sleeve moves to compress on the substrate when the collar is slid over the sleeve.

D7. The method according to any preceding D embodiment, wherein the sleeve defines a plurality of openings to receive a plurality of substrates, wherein the sleeve maintains spatial orientation of the plurality of substrates.

D8. The method according to any preceding D embodiment, wherein the heating cartridge further defines a first filament port in fluid communication with the interior cavity to receive the first filament and a second filament port in fluid communication with the interior cavity to receive a second filament, wherein the method further comprises:

feeding the second filament through the second filament port into the interior cavity; and

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melting the second filament in the interior cavity to form the printed jacket comprising material from the second filament.

D9. The method according to any preceding D embodiment, further comprising changing a feeding force applied to at least one of the first filament and the second filament to change a ratio of material in the printed jacket over a longitudinal distance.

D10. The method according to any preceding D embodiment, wherein the first filament has a Shore durometer less than or equal to 90A, 80A, 70A, 80D, 72D, 70D, 60D, 50D, 40D, or 35D.

Thus, various embodiments of ADDITIVE MANUFACTURING FOR MEDICAL DEVICES are disclosed. It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

In one or more examples, the described techniques may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include non-transitory computer-readable media, which corresponds to a tangible medium such as data storage media (e.g., RAM, ROM, EEPROM, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer).

Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term "processor" as used herein may refer to any of the foregoing structure or any other physical structure suitable for implementation of the described techniques. Also, the techniques could be fully implemented in one or more circuits or logic elements.

All references and publications cited herein are expressly incorporated herein by reference in their entirety for all purposes, except to the extent any aspect directly contradicts this disclosure.

Unless otherwise indicated, all numbers expressing feature sizes, amounts, and physical properties used in the specification and claims may be understood as being modified either by the term "exactly" or "about." Accordingly, unless indicated to the contrary, the numerical parameters set forth in the foregoing specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein or, for example, within typical ranges of experimental error.

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As used herein, the term “configured to” may be used interchangeably with the terms “adapted to” or “structured to” unless the content of this disclosure clearly dictates otherwise.

The singular forms “a,” “an,” and “the” encompass 5 embodiments having plural referents unless its context clearly dictates otherwise.

As used herein, “have,” “having,” “include,” “including,” “comprise,” “comprising” or the like are used in their open-ended sense, and generally mean “including, but not limited to.” It will be understood that “consisting essentially of,” “consisting of,” and the like are subsumed in “comprising,” and the like.

Reference to “one embodiment,” “an embodiment,” “certain embodiments,” or “some embodiments,” etc., means 15 that a particular feature, configuration, composition, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of such phrases in various places throughout are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, configurations, compositions, or characteristics may be combined in any suitable manner in one or more embodiments.

The words “preferred” and “preferably” refer to embodiments of the disclosure that may afford certain benefits, 25 under certain circumstances. However, other embodiments may also be preferred, under the same or other circumstances. Furthermore, the recitation of one or more preferred embodiments does not imply that other embodiments are not useful and is not intended to exclude other embodiments 30 from the scope of the disclosure.

What is claimed:

1. An additive manufacturing system for producing a 35 medical catheter or lead, the system comprising:

a heating cartridge extending from a proximal side to a distal side and comprising a substrate inlet port at the proximal side and a substrate outlet port at the distal side, the heating cartridge defining an interior volume and a substrate channel extending through the interior 40 volume from the proximal side to the distal side, wherein the heating cartridge defines a first filament port in fluid communication with the interior volume to receive a first filament;

a heating element thermally coupled to the heating cartridge to heat the interior volume;

a filament handling system comprising one or more motors to feed at least the first filament through the first filament port into the interior volume;

a substrate handling system comprising:

a clamp configured to secure a distal portion of an elongate substrate, wherein the substrate is positioned to pass through the substrate channel when secured by the clamp, wherein the clamp comprises a sleeve and a collar, and wherein the elongate 55 substrate extends through the sleeve and the collar is configured to compress the sleeve on the elongate substrate, and

one or more motors to move one or both of the substrate when secured by the clamp and the heating cartridge 60 relative to one another; and

a controller operably coupled to the heating element, one or more motors of the filament handling system, and one or more motors of the substrate handling system, the controller configured to:

activate the heating element to melt any portion of the first filament in the interior volume;

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control the one or more motors of the filament handling system to selectively control the feeding of the first filament into the interior volume; and

control one or more motors of the substrate handling system to move one or both of the substrate and the heating cartridge relative to one another in at least a longitudinal direction to form a printed jacket around the substrate, wherein the printed jacket comprises material from the first filament.

2. The system of claim 1, wherein the heating cartridge comprises an inlet die, a heating block, and an outlet die, wherein the inlet die defines a plurality of openings configured to receive a plurality of substrates, wherein the inlet die maintains spatial orientation of the plurality of substrates.

3. The system of claim 1, wherein the collar defines an inner hole to receive the sleeve, and wherein the collar is configured to slide along the sleeve to move at least a portion of the sleeve to compress the sleeve against the substrate and restrict movement of the sleeve relative to the substrate.

4. The system of claim 1, wherein the sleeve comprises a deformable portion and a rigid portion arranged along a longitudinal axis, wherein the collar is configured to slide onto the deformable portion and move the deformable portion to compress the sleeve onto the substrate such that movement of the substrate relative to the sleeve is restricted.

5. The system of claim 1, wherein the sleeve defines an increasing taper from an end of the sleeve, wherein the collar defines an inner hole having a fixed diameter, wherein the collar is configured to slide over the sleeve from the end of the sleeve and compress the sleeve as the collar moves further onto the sleeve.

6. The system of claim 1, wherein the sleeve defines one or more slots such that at least a portion of the sleeve moves to compress on the substrate when the collar is slid over the sleeve.

7. The system of claim 1, wherein the sleeve defines a plurality of openings to receive a plurality of substrates, wherein the sleeve maintains spatial orientation of the plurality of substrates.

8. The system of claim 1, wherein the heating cartridge further defines a first filament port in fluid communication with the interior volume to receive the first filament and a second filament port in fluid communication with the interior volume to receive a second filament, wherein the one or more motors of the filament handling system is configured to feed the second filament through the second filament port into the interior volume, wherein the controller is configured to:

activate the heating element to melt any portion of the second filament in the interior volume, and

control the one or more motors of the filament handling system to selectively control the feeding of the second filament into the interior volume, wherein the printed jacket comprises material from the second filament.

9. The system of claim 8, wherein the controller is configured to change a feeding force applied to at least one of the first filament and the second filament to change a ratio of material in the printed jacket over a longitudinal distance.

10. The system of claim 1, wherein the first filament has a Shore durometer less than or equal to 90A, 80A, 70A, 80D, 72D, 70D, 60D, 50D, 40D, or 35D.

11. The system of claim 1, wherein the substrate outlet port of the heating cartridge defines an interior shoulder, wherein the controller is configured to vary at least one of: a longitudinal speed of the substrate relative to the heating cartridge, a feeding force applied to the first filament, and an

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amount of heat provided by the heating element to change a thickness of the printed jacket over a longitudinal distance.

**12.** A method for additive manufacturing of a medical catheter or lead, the method comprising:

- securing a substrate via a clamp, wherein the clamp comprises a sleeve and a collar, and wherein the substrate extends through the sleeve and the collar is configured to compress the sleeve on the substrate;
- feeding the substrate through a substrate channel in a heating cartridge, the substrate channel in fluid communication with an interior cavity of the heating cartridge;
- feeding at least a first filament through a filament port into the interior cavity;
- melting the first filament in the interior cavity; and
- moving the heating cartridge relative to the substrate at least in a longitudinal direction to form a printed jacket comprising material from at least the first filament.

**13.** The method of claim **12**, wherein the heating cartridge comprises an inlet die, a heating block, and an outlet die, wherein the inlet die defines a plurality of openings configured to receive a plurality of substrates, wherein the inlet die maintains spatial orientation of the plurality of substrates.

**14.** The method of claim **12**, wherein the collar defines an inner hole to receive the sleeve, and wherein securing the substrate via the clamp comprises sliding the collar along the sleeve to move at least a portion of the sleeve to compress the sleeve against the substrate and restrict movement of the sleeve relative to the substrate.

**15.** The method of claim **12**, wherein the sleeve comprises a deformable portion and a rigid portion arranged along a longitudinal axis, wherein securing the substrate via the clamp comprises sliding the collar onto the deformable portion and moving the deformable portion to compress the sleeve onto the substrate such that movement of the substrate relative to the sleeve is restricted.

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**16.** The method of claim **12**, wherein the sleeve defines an increasing taper from an end of the sleeve, wherein the collar defines an inner hole having a fixed diameter, wherein securing the substrate via the clamp comprises sliding the collar over the sleeve from the end of the sleeve and compressing the sleeve as the collar moves further onto the sleeve.

**17.** The method of claim **12**, wherein the sleeve defines one or more slots such that at least a portion of the sleeve moves to compress on the substrate when the collar is slid over the sleeve.

**18.** The method of claim **12**, wherein the sleeve defines a plurality of openings to receive a plurality of substrates, wherein the sleeve maintains spatial orientation of the plurality of substrates.

**19.** The method of claim **12**, wherein the heating cartridge further defines a first filament port in fluid communication with the interior cavity to receive the first filament and a second filament port in fluid communication with the interior cavity to receive a second filament, wherein the method further comprises:

- feeding the second filament through the second filament port into the interior cavity; and
- melting the second filament in the interior cavity to form the printed jacket comprising material from the second filament.

**20.** The method of claim **19**, further comprising changing a feeding force applied to at least one of the first filament and the second filament to change a ratio of material in the printed jacket over a longitudinal distance.

**21.** The method of claim **12**, further comprising positioning an electrical subassembly on the substrate, wherein the printed jacket covers at least a portion of the electrical subassembly and is spaced apart from the electrode ring; and removing the substrate from the printed jacket.

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